Module-1

Introduction to Artificial Intelligence

Introduction to Artificial intelligence: What is AI, History of AI, Types of AI, Components of Intelligence, Agent and Environment, Informed Search Algorithms: Pure Heuristic, Best first, A*, AO* Algorithms.

1. What is Artificial Intelligence?

Artificial Intelligence (AI), simply means the intelligence demonstrated by machines that help them to mimic the actions of humans. AI simulates natural intelligence in machines that are programmed to learn from experiences, adjust to new inputs and perform human-like tasks.

1.1 Advantages and Disadvantages of Artificial Intelligence

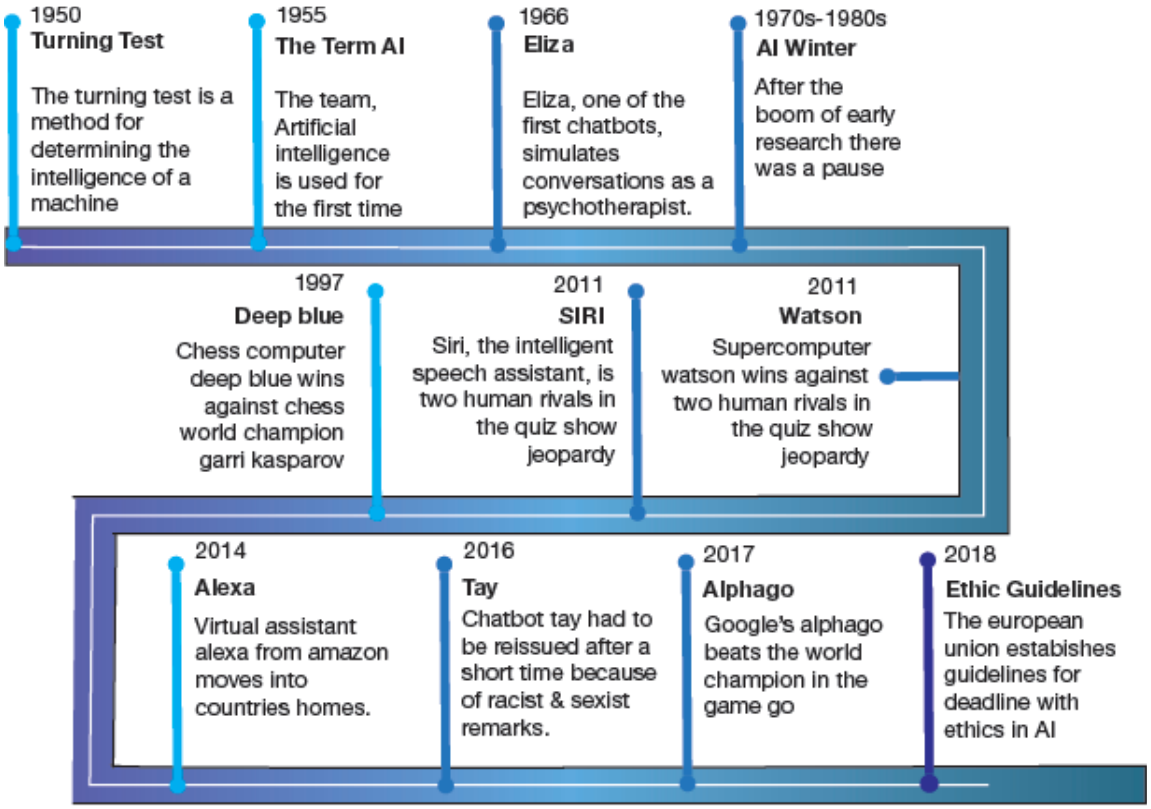
Advantages:

1. Performs well on tasks that uses detailed data.
2. Takes less time to perform tasks that needs to process huge volumes of data.
3. Generates consistent and accurate results.
4. Can be used 24 X 7.
5. Optimizes tasks by better utilizing resources.

Disadvantages:

1. Involves more cost.
2. Technical expertise required to develop and use AI applications.
3. Lack of trained professionals.
4. Incomplete or inaccurate data may result in disastrous results.
5. Lacks the capability to generalize tasks.

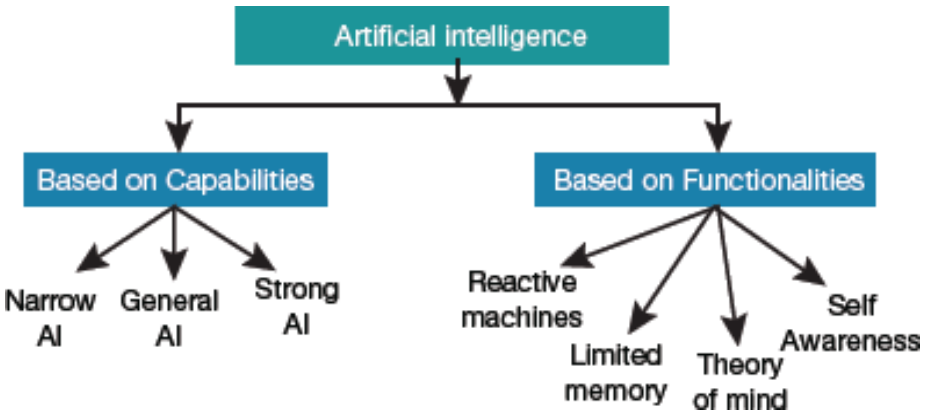
2. History of Artificial Intelligence



- In **1952**, Arthur Samuel developed a self-learning program to play checkers.
- In **1954**, the IBM computer could translate 60 carefully selected Russian sentences into English.
- In **1956**, John McCarthy coined the term 'artificial intelligence'. The scope and goals of AI were discussed in a conference later.
- In **1997**, IBM's Deep Blue machine could beat world chess champion, Gary Kasparov.
- In **2005**, STANLEY, a self-driving car, won the DARPA Grand Challenge. Even the US military started investing in autonomous robots like Boston Dynamics' 'Big Dog' and iRobot's 'PackBot.'

- In **2008**, Google did a remarkable work in speech recognition and introduced the feature in its iPhone app.
- In **2011**, Apple released Siri, an AI-powered virtual assistant that could beused through its iOS operating system.
- In **2012**, Andrew Ng, founder of the Google Brain Deep Learning project, fed 10 million YouTube videos as a training set to a neural network and used deep learning algorithms to enable the neural network to recognize a cat without being told what a cat is.
- In **1958**, John McCarthy developed the AI programming language, Lisp. He also published a paper discussing a complete AI system that could learn from experience as effectively as humans do.
- In **1959**, Allen Newell, Herbert Simon and J.C. Shaw developed the General Problem Solver (GPS), a program designed to imitate human problem solving.
- In **2016** Google DeepMind's AlphaGo defeated world champion Go player Lee Sedol. Go is an ancient Chinese game and winning this game was a major breakthrough in the field of AI.
- In the same year, the first 'robot citizen', a humanoid robot, Sophia was created that could perform facial recognition, verbal communication and facial expression.
- In **2018**, Google released NLP engine BERT that reduced barriers in translation and understanding by machine learning applications. In the same year, Waymo One service was launched that allowed users to request a pick-up from one of the company.
- In **2020**, Baidu released its Linear Fold AI algorithm to develop a vaccine during the early stages of the SARS-CoV-2 pandemic. The algorithm could predict the RNA sequence of t.

3. Types of Artificial Intelligence



1. **Weak AI**:

Weak AI, also known as narrow AI,

is specifically designed to perform a specific type of task. For example, Siri and Alexa are weak AI systems. These systems are already trained with appropriate responses to classify things accordingly. When you instruct Alexa to play a song, it responds by playing that song.

In fact, majority of AI applications that we use today (predicting weather, stock prices, optimizing business, etc.) come under this category. Weak AI systems operate within a limited context and are the most successful realization of artificial intelligence to date. Application of narrow AI has resulted in significant societal benefits. Google search, Image recognition software, self-driving cars and IBM's Watson are some examples of such systems.

2. **Strong AI**:

Strong AI, also known as artificial general intelligence or artificial super intelligence (ASI) or superintelligence, makes full attempt to resemble the human brain. It utilizes cognitive skills and fuzzy logic to perform tasks for which it had not been trained earlier. Such a system needs capability for visual perception, speech recognition, decision-making and translations between languages.

As of now, we see general AI only in sci-fi movies, where machines emulate human intelligence and think strategically to perform complex tasks.Many people fear that intelligent robots would overrun humanity, but experts says that we need not worry about it as it is not likely to happen in the near future.

3. Reactive Machines:

Reactive machines are very basic machines with no memory to store and thus use past experiences to determine future actions. They just perceive the world and react to it.

IBM's Deep Blue, which defeated chess grandmaster Kasporov, is such an example. The AI system sees the pieces on a chess board and responds without referring to any of its prior experiences.

Since it lacks capabilities to store past experiences, a reactive machine cannot improve with practice.

Google's AlphaGo is another example of a game-playing reactive machine. The application evaluates future moves using neural network. This gives it an edge over Deep Blue game.

4. Limited Memory:

Limited memory machines retain data but only for a short period of time and can use it only for a limited time. They cannot keep adding data permanently to a library of their experiences. Such systems are usually used in autonomous vehicles since they need not store data (like *recent speed of nearby cars, distance between cars, the speed limit, etc.)* that can help them navigate roads.

AlphaGo also bested world-class competitors of the game, defeating champion Go player Lee Sedol in 2016.

No doubt, limited memory AI is more complex and is used for better possibilities than reactive machines. These machines continuously train a model to analyse and utilize new data. The model also improves using feedback received from humans or the environment and stored as data.

5. Theory of Mind:

The theory of mind focuses on imitating the human brain. For this, it form representations about the world, starting from thoughts, emotions and memories. However, such systems are only theoretical as of now but they may become reality very soon.

These machines must understand thoughts and emotions that affect the behaviour of one's self. Theory of Mind machines make decisions keeping in view the feelings experienced through self-reflection and determination.

6. Self-Awareness:

Once Theory of Mind is established, then the next step would be incorporating self-awareness in AI systems. Self-awareness in machines enable them to possess human-level consciousness, understand the current state—its own existence in the world and use that information to deduce others feelings.

Self-AI systems will be able to understand what others may need. They could interpret user's feelings by learning not only what they communicate to them but also how they communicate it.

4. Components of Intelligence

- Reasoning
- Learning
- Problem solving
- Perception
- Linguistic intelligence

1. Reasoning

It is the set of processes that is used in making decisions and prediction.There are two types of reasoning—inductive and deductive.

Inductive Reasoning	Deductive Reasoning
It uses specific observations to make broad general statements.	It starts with a general statement to obtain a specific, logical conclusion.
Even if all observations are true, there are still chances that the conclusion can be false.	If something is true for a particular class of things in general, it is also true for all members of that class.
Example: 'Manvi is a topper. Manvi is studious. Therefore, all toppers are studious.'	Example: 'Students who scored at least 80% are eligible for placement. Manan scored 90%. Therefore, Manan is an eligible candidate.'

2. **Learning**

It is a process of gaining knowledge or skill by studying, practicing, being taught or experiencing something. Learning is an ability that is possessed by humans, animals and AI-enabled systems to improve the awareness of the subjects of the study.

Learning can be categorized as follows.

Auditory learning : promotes learning by listening and hearing. For example, students hear recorded lectures to understand a particular concept.

Episodic learning: promotes learning by remembering the sequences of events that one has witnessed or experienced. This is linear and orderly.

Motor learning: helps to learn by precise movement of muscles. For example, picking objects.

Observational learning: triggers learning by watching and imitating others. For example, children often learn by mimicking their parents.

Perceptual learning: is done by recognizing stimuli that one has seen before. For example, identifying and classifying objects and situations.

Relational learning: involves learning to differentiate among various stimuli on the basis of relational properties, rather than absolute properties.

3. **Problem Solving**

It is the process in which one tries to find the desired solution in the present

situation by following the path which is blocked by known or unknown hurdles. Problem solving uses **decision-making to** select the best suitable alternative to reach the desired goal.

4. **Perception**

It is the process of acquiring, interpreting, selecting and organizing sensory information. While humans use sensory organs to perceive their environment, AI systems use data acquired by the sensors to do the same.

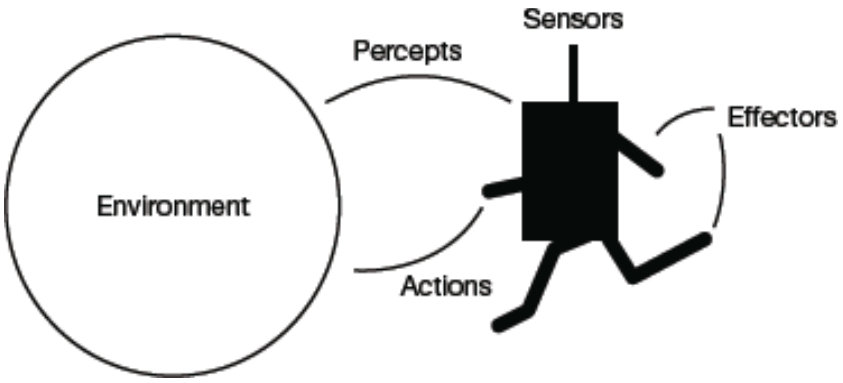
5. **Linguistic Intelligence**

It is used in in interpersonal communication and defines one's ability to use, comprehend, speak and write the verbal and written language.

5. Agent and Environment

Artificial intelligence is all about studying rational agents, which may include anything that makes decisions. For example, a person, firm machine or software that performs action(s) with the best outcome after considering past and current percepts at a given instance are examples of agents. These agents work in their

environment and may interact with other agents in that environment. Thus, we can say that an AI system comprises an **agent and its environment**.



In an artificially intelligent system, the agents act in their environment which may contain other agents.
This **agent** can be anything that is capable of perceiving its environment. For this, it may have **sensors.**

Agents also have effectors that help them to act upon their environment**.**

The different types of agents in an AI system include the following:

1. H**uman agent:** A human agent has sensory organs (like eyes, ears, nose, tongue and skin) that acts as sensors, and other organs such as hands, legs, mouth as effectors for taking actions.

2. R**obotic agent:** A robotic agent uses cameras and infrared range finders as sensors, and various motors and actuators as effectors.

3. **Software agent:** Such an agent uses bit strings as its programs and actions.

6. Types of Agents

1.Simple Reflex Agents

Simple reflex agents choose actions based only on the current percept. They are rational only if they make a correct decision based on the current precept. Their environment is completely observable.

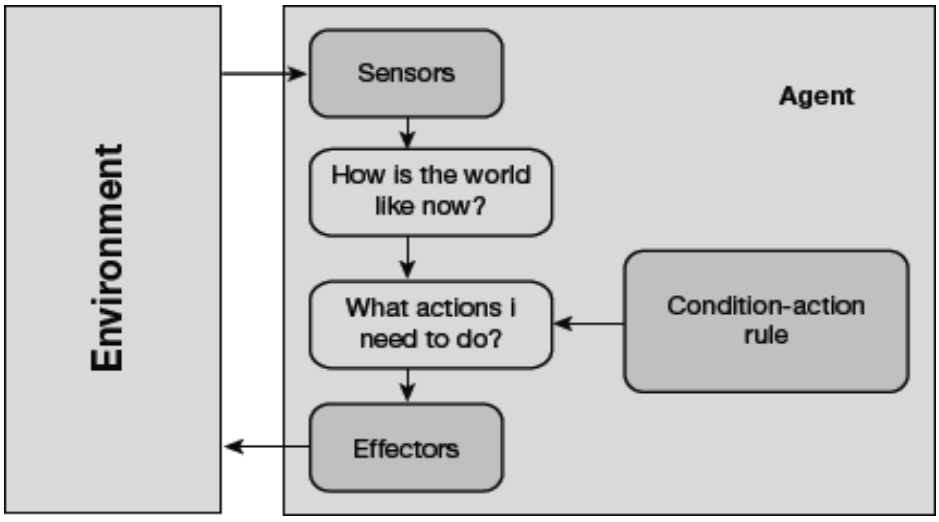
They work using **condition-action rule** that maps a state (condition) to an action. If the condition is true, then the action is taken, else not. However,

the problem with a simple reflex agent is that they succeed only when the environment is fully observable.

1. Possess very limited intelligence
2. Has knowledge of any state other than the current state.
3. Any change in the environment will eventually end up updating the collection of rules.

2.Model-Based Reflex Agents

Model reflex agents use a model of the world to choose their actions. For this, they need to maintain an internal state. Here, **model** means knowledge about 'how things happen in the world'.



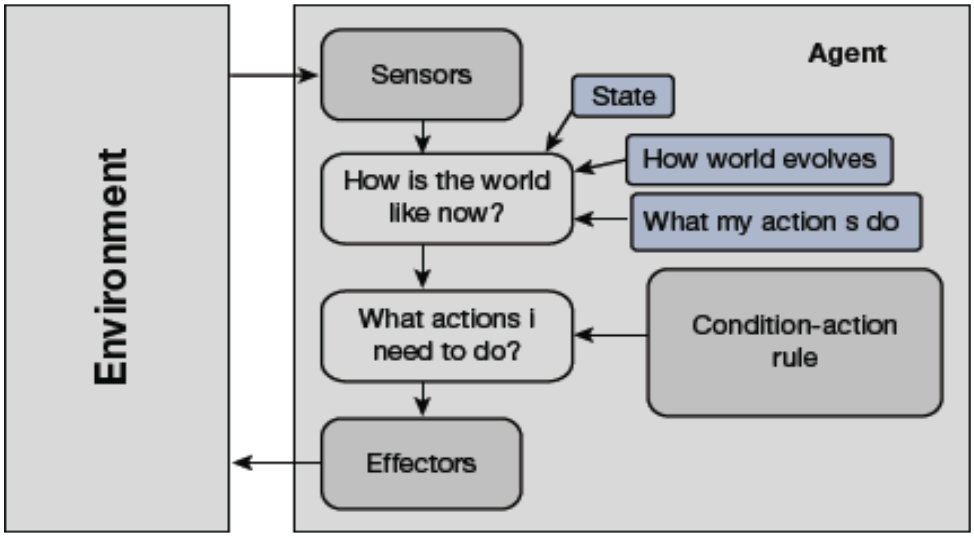
Internal state represents unobserved aspects of the current state depending on percept history.

To update the state, information is required to understand,

- How the world evolves and
- How the agent's actions affect the world.

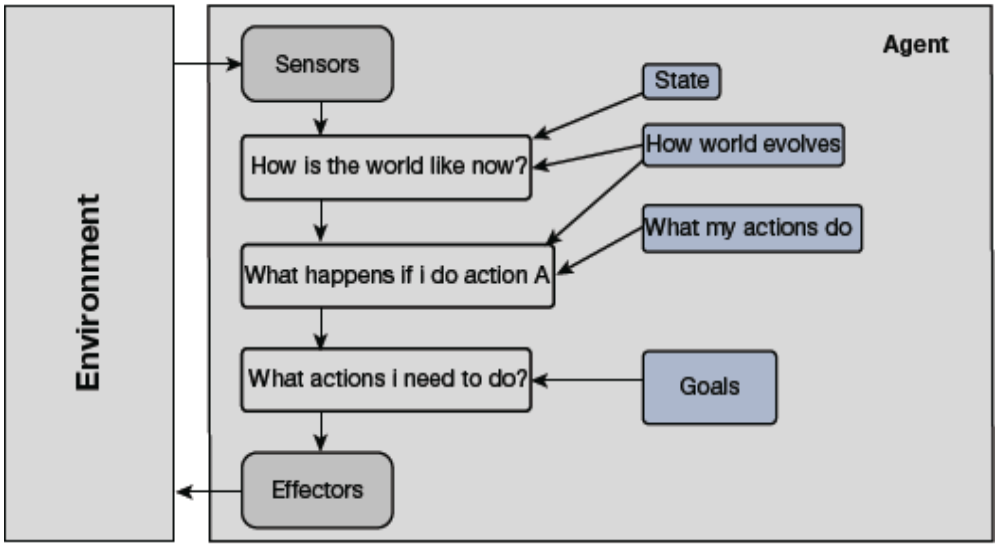
3.Goal-Based Agents

Goal means a description of desirable situations. Goal-based agents choose their actions to achieve goals. This provides more flexibility than reflex agent as the knowledge supporting a decision isexplicitly modeled and permits any sort of modifications.



4.Utility-Based Agents

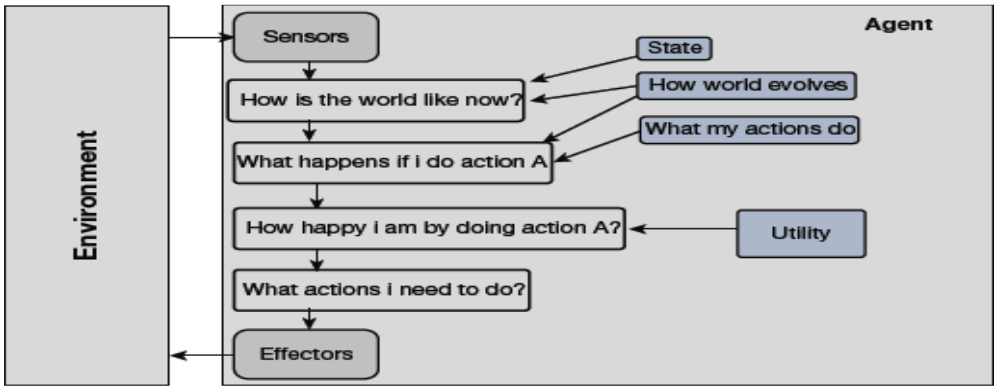
Many a time, goals are either conflicting or are difficult to achieve. In suchsituations, utility-based agents can be used to achieve goals that are more important than others. These agents choose actions based on a preference (utility) for each state.



However, at times, achieving the desired goal is not enough and happiness of the agent is also important. In such a scenario, utility describes how**'happy'** the agent is. A utility agent chooses the action that maximizes the expected utility, that is, the associated degree of happiness.

5.Learning Agent

A learning agent learns from its past experiences. It is designed to possess learning capabilities. Initially, it starts working with basic knowledge and then acts and adapts automatically through learning.



Such an agent has four main conceptual components, as follows:

Learning element, which is responsible for making improvements by learning from the environment

Critic is **t**he element that provides feedback to the learning element, describing how well the agent is doing with respect to a fixed performance standard.

Performance element chooses a particular external action to be taken.

Problem generator suggests actions that result in new and informative experiences.

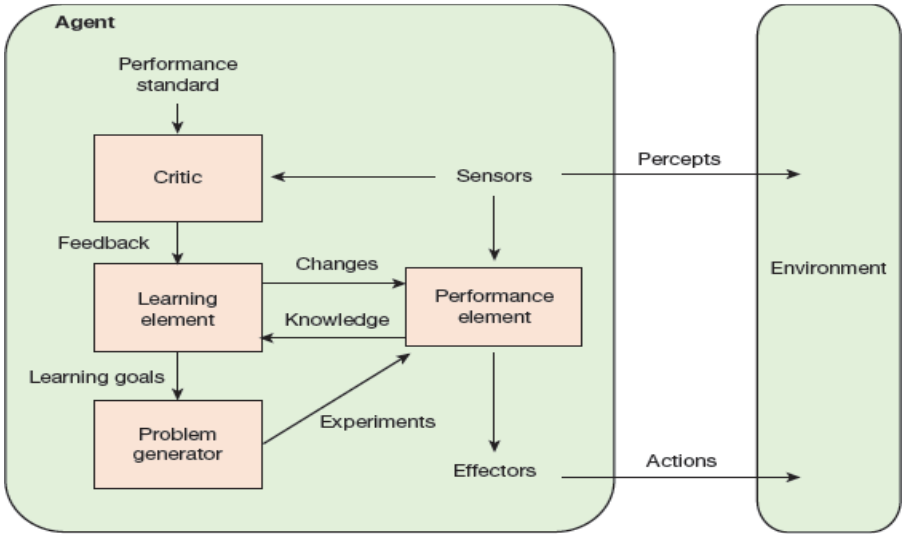


Fig: Citric

7. Types of Environments

As stated earlier, environment is the place where the agent would work. It rewards and specifies states and actions of the agents. Although environment in an AI system is quite vast, we can identify a small number of dimensions to categorize them as given below.

A. Discrete/Continuous

If an environment has a limited number of distinct, clearly defined, states, then it is said to be discrete. For example, a software agent designed to play chess game works in a discrete environment as there are limited moves that a player can make. However, a self-driving car operates in a continuous environment.

While a discrete environment has a finite number of percepts and actions that can be performed within it, a continuous environment has no such constraints.

B. Known Vs Unknown

Known and unknown are based on an agent's state of knowledge to perform an action and they are not actually a feature of an environment. In a known environment, the agent knows the results for all actions but in an unknown environment, the agent has to learn how it should work to perform an action.

Usually, a known environment is partially observable and an unknown environment is fully observable.

C. Observable/Partially Observable

If an agent can determine the complete state of the environment at each time point from the percepts, then it is said to be observable; otherwise, it is only partially observable. Correspondingly, in an unobservable environment, the agent has no sensors. In a fully observable environment, the complete state of the environment *relevant to the choice of action* of the agent can be captured using its sensors.

D. Static/Dynamic

A static environment does not change when an agent is acting, but a dynamic environment does. Agents in static environments, therefore, do not take into account the state of the environment during the action and are thus easy to deal with. However, in a dynamic environment, agents must consider the world while

performing each action. For example, self-driving cars work in a dynamic environment, but games like crossword puzzles and chess operate in a static one.

Between these two extremes, we can also have **semi-dynamic environments** which do not change with the passage of time but the agent's performance score does.

E. Single Agent/Multiple Agents

The environment may contain other agents which may or may not be of the same kind of the agent. For example, vacuum cleaning environment is a single agent but a chess game is a two-agent environment.

In a multi-agent environment, the performance measure of one agent depends on the action performed by another agent. A multi-agent environment may be competitive or cooperative.

F. Accessible/Inaccessible

In an accessible environment, the sensory apparatus of the agent has access to the complete state of the environment. This means that it can obtain complete and accurate information about the state's environment, but this is not so in case of an inaccessible environment.

For example, an empty room whose state can be defined by its temperature is an example of an accessible environment, whereas information about an event on the Earth is an inaccessible environment.

G. Deterministic/Non-deterministic

In a deterministic environment the next state of the environment can be easily determined by the current state as well as the actions of the agent. This is not true for a non-deterministic environment. A n**on-deterministic** environment description maximizes an agent's performance for all possible outcome of its actions. Note that in a deterministic, fully observable environment, the agent does worry about uncertainty.

In a **Stochastic** environment, uncertainty about outcomes is quantified in terms of **probabilities**.

H. Episodic/Non-episodic

In an episodic environment, each episode has an agent perceiving and then acting. The quality of its action depends just on the episode itself. Subsequent episodes do not depend on the actions performed in the previous episodes.

I. Search

Algorithms

We have read that artificial Intelligence is the study of building rational agents. These agents make use of some or the other search algorithms in the background to achieve their tasks. For example, some single-player games like tile games, Sudoku, crossword, etc., use search algorithms to deduce a particular position in the game.

A search problem consists of the following:

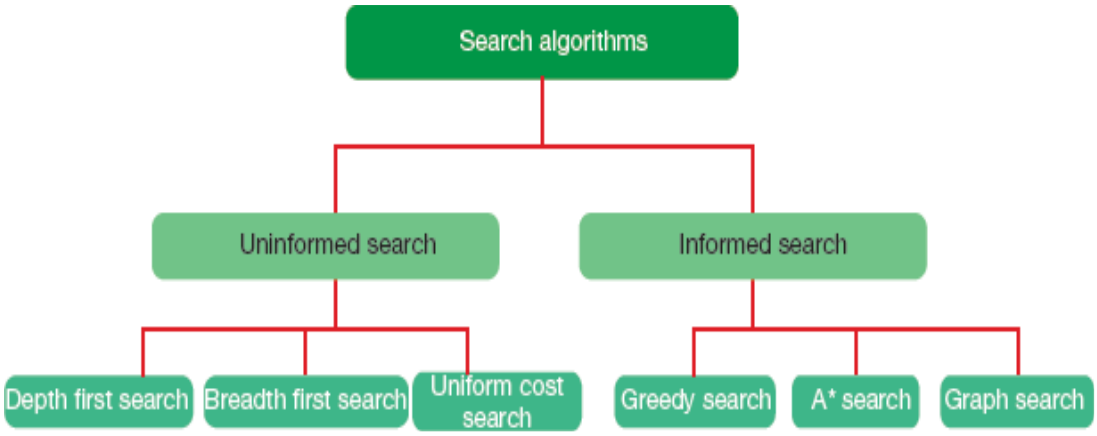
1. **State space:** is the set of all possible states that can be attained by an agent.

2. **Start state:** is the state from where the searching is done.

3. **Goal test:** is a function that checks if the current state is the goal state or not.

4. **Solution:** to a search problem is a sequence of actions (also known as **plan)** that transforms the start state to the goal state. This plan is realized using search algorithms.

7.1 Types of Search Algorithms



1. Informed Search Algorithms:

The informed search algorithm which is extensively used in a large space is also known as heuristic search as it uses a heuristic function. The heuristic function takes the current state of the agent as its input and estimates how close the agent is from the goal. Though the heuristic method may not always give the best solution, it is, however, guaranteed to find a good solution in reasonable time.

A heuristic function h(n), calculates the cost of an optimal path between the pair of states. The value of the heuristic function is always positive. We can write,

h(n) <= h* (n)

where h(n) is heuristic cost, and h (n) is the estimated cost. This means that the heuristic cost should be less than or equal to the estimated cost**.**

7.1.1 Pure Heuristic Search

To solve large problems with a large number of possible states, problem specific knowledge needs to be added to increase the efficiency of search algorithms. In such a situation, heuristic search algorithm performs well.

Heuristic search is the simplest form of heuristic search algorithms that expands nodes based on their heuristic value h(n). The algorithm maintains two lists—OPEN and CLOSED. All nodes that have already been expanded are placed in the CLOSED list, while others that have not been expanded are in the OPEN list.

a. Best-First Search Algorithm (Greedy Search)

Greedy best-first search algorithm combines depth-first search and breadth first search algorithms and always selects the path which appears best at that moment. It uses the heuristic function and best-first search at each step to choose the most promising node. The most promising node is the one that

is closest to the goal node. The greedy best first algorithm is implemented by the priority queue.

Best first search algorithm steps:

Step 1: Insert the starting node in the OPEN list.

Step 2: Stop and return failure if the OPEN list is empty,

Step 3: From the OPEN list, remove the node having the lowest value of h(n), and insert it in the CLOSED list.

Step 4: Expand the node (removed in Step 3) and generate its successors.

Step 5: Check each successor to find if it is the goal node or not. If any successor node is found to be the goal node, then return success and terminate the search, else go to Step 6.

Step 6: Check if each successor node is in either OPEN or CLOSED list. If the node is not present in any of the lists, then add it to the OPEN list.

Step 7: Go to Step 2.

Advantages

1. It takes advantage of both BFS and DFS algorithms.

2. Best first search algorithm is more efficient than BFS and DFS algorithms.

Disadvantages

1. In some worst-case scenarios, best first search can behave as an unguided depth-first search.

2. Like DFS, best first search algorithm can get stuck in a loop.

3. Best first search algorithm is not an optimal algorithm.

Example: Consider the tree given below and traverse it using greedy best first search algorithm.

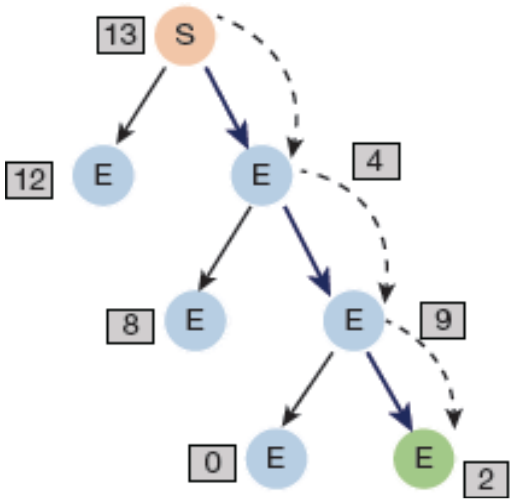


Table shows the heuristic estimates:

Node	h(n)	Node	h(n)	Node	h(n)
A	11	E	4	I, J	3
B	5	F	2	S	15
C, D	9	H	7	G	0

Solution: Expand the node S and put it in the CLOSED list. Generate its successors and place them in the OPEN list.

Open [A, B], Closed [S]

Iteration 1: Since h(n) for node B is less than that of node A, expand B and place it in CLOSED list. Now only node A is in the OPEN list. Open [A], Closed [S, B]

Iteration 2: Generate successors of node B and place them in the OPEN list. Open [E, F, A], Closed [S, B].

Since node F has the lowest heuristic value among all nodes in the OPEN list, we will expand node F. Now, Open [E, A], Closed [S, B, F].

Iteration 3: Generate successors of node F and place them in the OPEN list. So, Open [I, G, E, A], Closed [S, B, F].

Now, generate succors of node F and place them in OPEN list. Since one of the successors of node F is the goal node, the algorithm returns success and terminates. The path is given by the nodes present in the CLOSED list.

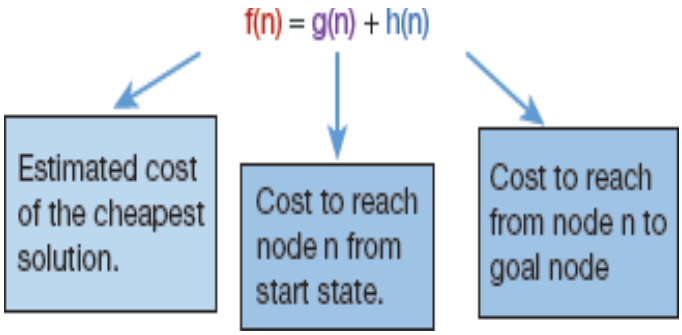
Open [I, E, A], Closed [S, B, F, G]

Hence, the final solution path will be: **S----> B----->F----> G**

The worst-case time complexity and space complexity of greedy best first search is O(b**m**) where, m is the maximum depth of the search space.

b. A* Search Algorithm

The A search algorithm, best-known form of best first search, uses heuristic function h(n), and g(n) which specifies the cost to reach the node n from the start state. It solves the searching problem efficiently by expanding less search tree to find shortest path and give an optimal result faster. That is, it avoids expanding paths that are already expensive, but expands most promising paths first.



The A search algorithm, uses search heuristic as well as the cost to reach the node. Sum of both the costs gives a number known as a **fitness number**.

Step1: Put the starting node in the OPEN list.

Step 2: If the OPEN list is empty then return failure and stop.

Step 3: From the OPEN list, choose the node having minimum value of f(n). If this node is the goal node, then return success and stop.

Step 4: Expand the node selected in Step 3 and generate its successors. Put the node in the closed list. For each successor, check if it is present in the OPEN or CLOSED list. If it is not present in either then compute its f(n)and place it into Open list.

Step 5: Else if the successor node is already in OPEN or CLOSED list, then attach it to the back pointer reflecting the lowest g(n') value.

Step 6: Return to **Step 2**.

Advantages

1. It is better than all other search algorithms.

2. A search algorithm is optimal and complete.

3. The algorithm can be used to solve very complex problems.

4. It is used by many games and web-based maps to find the shortest path efficiently.

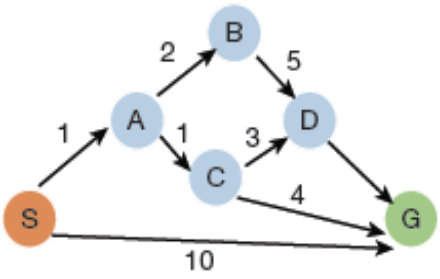
Disadvantages

1. It may not always produce the shortest path since it is based on heuristics and approximation.

2. A search algorithm is more complex than other search algorithms

3. It uses more memory as it keeps all generated nodes in the memory. This makes it difficult to use for large-scale problems.

Example: Consider the graph given below and traverse it using the A algorithm. Note than h(n) will be calculated by adding h(n) and the cost of reaching that node.



State	h(n)
S	5
A	3
B	4
C	2
D	6
G	0

Solution:

Initially, we start with node S having h(n) = 5. Therefore, {(S, 5)}

Iteration1: From node S, we can reach node A and node G.

h(n) of node A is 3 and the cost to reach node A from node S is 1. Therefore, f(n) of node A = 4. Similarly, node G has h(n) = 0 and the cost to reach node

G from node S is 10. Therefore, f(n) of node G = 10.

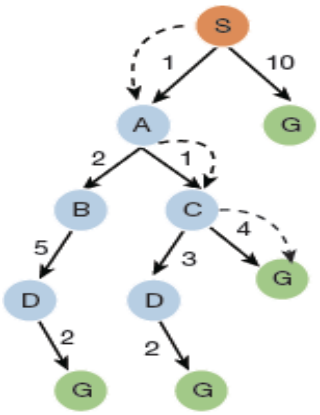
{(S--> A, 4), (S-->G, 10)}

Iteration2: Since node A has less f(n) as compared to node G, we will expand node A.

h(n) of node B is 4 and the cost to reach node B from node S is 3.

Therefore, f(n) of node B = 7. Similarly, node C has h(n) = 2 and the cost to reach node C from node S is 2. Therefore, f(n) of node C = 2.

{(S--> A-->C, 4), (S--> A-->B, 7), (S-->G, 10)}



Iteration3: Since node C has less f(n) as compared to other nodes, we will expand node C.

h(n) of node D is 6 and the cost to reach node D from node S is 5.

Therefore, f(n) of node D = 1. Similarly, node G has h(n) = 0 and the cost to reach node G from node S is 6. Therefore, f(n) of node G = 2.

{(S--> A-->C--->G, 6), (S--> A-->C--->D, 11), (S--> A-->B, 7), (S-->G, 10)}

Iteration 4: Since G is the goal node, the algorithm will return success and terminate giving the final path as, **S--->A--->C--->G** with cost 6.

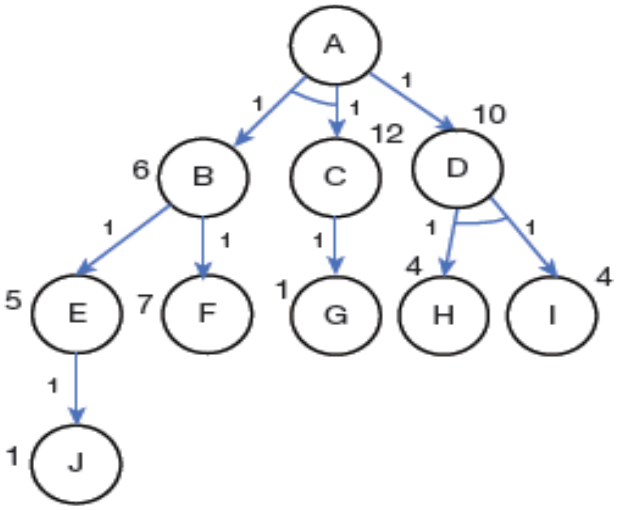
c. AO* Algorithm

- AO algorithm is based on problem decomposition in which a problem is broken down problem into smaller pieces.Therefore, AO* algorithm can be applied only on those algorithms that can be divided into a set of sub problems. Each sub problem can be solved eparately. The final result is obtained by combining all the solutions Usually, AND-OR graphs /trees are used for representing the solution. The decomposition of the problem generates AND arcs.

J. A* Vs AO*

- Both A and AO algorithms are informed search techniques and use heuristic values to solve the problem.
- Although both the algorithms guarantee solution, only A guarantees an optimal solution. By optimal, we mean the shortest path with the lowest cost. The AO algorithm does not always provide an optimal solution. This is because AO does not explore all the solution paths once it finds a solution.

- **Example:** Consider the tree given below and apply the AO* algorithm on it. Remember that



Remember that the AO algorithm always moves towards a lower cost value. The cost function, (F(n)= G (n) + H (n)), where, **H(n) is the**

heuristic or the estimated value of the node and G(n) is the actual cost or edge value.

Step 1: From node A, we can either choose D or B-C together. The cost of both the paths are calculated as,

F(A to D)= 1+10 = 11 and

F(A to B and C) = 1 + 1 + 6 +12 = 20

We will choose node D as the next node to traverse as it has a lower cost than path B-C.

Step 2: From D, we are left with only one choice H-I. Now, the cost of H-I from node D is calculated as,

F(D to H-I) = 1+1+ 4 +4 =10

Correspondingly, the cost of H-I from node A is 11 as,

F(A to D to F-E) = 1+10 =11

If the goal state is found on the path from A to H-I, then the other path remains unexplored.

Let us explore the other path from node A to B-C. We had calculated,

F(A to B and C) = 1 + 1 + 6 +12 = 20

Step 1: From B, there are two nodes can be reached- E and F.

F(B-E)= 5+1 = 6 and F(B-F) = 7 + 1 = 8

Step 2: Since cost from **F(B-F)** is more than that of the path B-E, the algorithm moves to node J.

F(E-J) = 1 +1 = 2, which is less than heuristic value 5. So, the new heuristic value from E is 2.

Step 3: We need to update the cost from E to B, which is now,

F(E-B)= 1+2 =3

Therefore, the new heuristic value of B is 3.

Step 4: We need to update the value of A as A is associated with both B and C.

But to update the heuristic value of A, we must know the value of node C. From C, node G can be reached with a cost of 1. So, the heuristic value of C is now 1 + 1 = 2.

Since cost form C to G = F(C-G) = 2 which is less than its heuristic value, the new heuristic value of C is 2.

Step 5: Now the updated from A- BC that is F(A-BC) = 1+1+2+3 = 7 which is less than F(A - D) = 11.

This shows choosing path A-BC is more cost effective and thus better path A-D.

However, this is possible if the algorithm explores this path as well. But the algorithm will not explore this path. We have just shown this path to demonstrate the weakness of the AO algorithm.

The AO* does not give optimal performance in all the cases.

Advantages

1. It gives optimal solution.
2. It can be used for both OR and AND graph when nodes are traversed in a particular order.

Disadvantages

1. It does not find the optimal solution in every case.
2. Its complexity is more than other algorithms.

Module 2: Knowledge Representation and Reasoning (KRR)

4.1 Introduction

Artificial Intelligence (AI) aims to develop machines that can think, learn, understand information, and make intelligent decisions similar to humans. Science fiction movies and novels often portray AI-powered robots capable of performing complex tasks intelligently.

However, creating machines that behave exactly like humans is challenging because human intelligence is developed through knowledge, experiences, memories, learning, and conscience.

To make machines intelligent, it is necessary to provide them with knowledge in a form that they can understand and use effectively. This process is called **Knowledge Representation**.

Knowledge and Intelligence

Knowledge is one of the most important components of intelligence.

An intelligent agent:

1. Senses its environment.
2. Uses knowledge about the environment.
3. Makes decisions.
4. Performs appropriate actions.

Relationship between Knowledge and Intelligence

Environment → Knowledge → Decision Making → Intelligent Action

Without knowledge:

- An agent cannot correctly understand its environment.
- It cannot reason effectively.
- It cannot make intelligent decisions.

Therefore,

Knowledge is the foundation of intelligence.

Example

Consider a self-driving car:

- It senses roads, vehicles, pedestrians, and traffic signals.
- Uses stored knowledge about driving rules.
- Makes decisions such as stopping, turning, or accelerating.
- Performs intelligent driving actions.

Without knowledge, the car would be unable to drive safely.

4.2 Knowledge Representation

Definition

Knowledge Representation (KR) is a branch of Artificial Intelligence that deals with representing knowledge in a form that a computer system can understand and use for reasoning and problem-solving.

Knowledge Representation and Reasoning (KRR) focuses on:

- How AI agents think.
- How knowledge is stored.
- How reasoning is performed.
- How intelligent behavior is produced.

Knowledge-Based System

A **Knowledge-Based System (KBS)** stores information about the real world in a structured form.

This enables computers to:

- Understand information.
- Draw conclusions.
- Solve problems.
- Communicate with humans.

Applications

Knowledge representation is used in:

1. Medical diagnosis systems.
2. Expert systems.

3. Natural language processing.
4. Robotics.
5. Self-driving vehicles.
6. Intelligent assistants.

Purpose of Knowledge Representation

The main objectives are:

1. Store Knowledge

Store information about the world.

2. Support Reasoning

Allow machines to think and derive conclusions.

3. Model Intelligent Behavior

Enable machines to behave intelligently.

4. Learn from Experience

Help machines improve performance using previous knowledge.

Important Point

Knowledge representation is **not merely storing data in a database**.

It goes beyond storage by enabling:

- Learning.
- Reasoning.
- Decision making.
- Problem solving.

Thus, machines can behave intelligently like humans.

4.2.1 What Knowledge Needs to be Represented?

Human intelligence consists of more than facts.

Humans use:

- Intuition

- Beliefs
- Intentions
- Judgments
- Common sense
- Experiences

To create intelligent AI systems, all relevant knowledge must be represented in a machine-understandable form.

The following types of knowledge need to be represented.

1. Object Knowledge

Definition

Information and facts about objects present in the environment.

Example

For a self-driving car:

- Vehicles
- Roads
- Traffic signals
- Pedestrians

Importance

The AI system must know:

- What the object is.
- Its characteristics.
- Its role in the environment.

2. Event Knowledge

Definition

Knowledge about actions or events occurring in the real world.

Example

In a self-driving car:

- A pedestrian crossing the road.
- A vehicle suddenly stopping.
- Applying brakes.

Importance

Helps the system understand changing situations and react appropriately.

3. Performance Knowledge

Definition

Knowledge describing **how actions should be performed**.

Example

A robot may know:

- How to pick an object.
- How to move safely.
- How to assemble components.

Importance

Describes behavior and execution methods.

4. Meta-Knowledge

Definition

Knowledge about knowledge.

It refers to understanding:

- What knowledge exists.
- Where it is stored.
- How it should be used.

Example

An expert system knowing which rule should be applied in a particular situation.

Importance

Improves reasoning efficiency.

5. Facts

Definition

Facts are statements that are true about the real world.

Examples

- Water boils at 100°C.
- Earth revolves around the Sun.
- Traffic light red means stop.

Importance

Facts form the foundation of reasoning.

6. Knowledge Base

Definition

A **Knowledge Base (KB)** is a collection of facts, rules, and information stored in a machine-readable form.

It is the most important component of a knowledge-based system.

Characteristics

- Stores knowledge systematically.
- Contains logical sentences.
- Supports reasoning.

Example

Medical Expert System Knowledge Base:

- Symptoms
- Diseases
- Treatment rules

4.2.2 What is Knowledge?

Definition

Knowledge is the fundamental element of human intelligence.

It helps people:

- Understand situations.
- Solve problems.
- Make decisions.
- Learn from experiences.

Knowledge is acquired through:

- Facts
- Data
- Experiences
- Observations
- Situations

Important Statement

A knowledgeable person performs actions more effectively and intelligently.

Types of Knowledge

Every individual possesses different forms of knowledge.

There are **five major types of knowledge**:

1. Meta Knowledge
2. Heuristic Knowledge
3. Procedural Knowledge
4. Declarative Knowledge
5. Structural Knowledge

1. Meta Knowledge

Definition

Meta Knowledge is **knowledge about knowledge**.

It describes:

- What knowledge is available.
- How knowledge should be used.

- When knowledge should be applied.

Example

An AI system knowing which rule is appropriate for solving a particular problem.

Importance

- Improves decision-making.
- Helps select relevant knowledge.
- Increases reasoning efficiency.

2. Heuristic Knowledge

Definition

Heuristic Knowledge is specialized knowledge gained from:

- Experience
- Expertise
- Practical observations

It is commonly known as a **Rule of Thumb**.

Characteristics

- Experience-based
- Practical
- Not guaranteed to be correct
- Often produces good solutions quickly

Example

A doctor's experience suggesting that certain symptoms may indicate a specific disease.

AI Example

Chess programs use heuristic rules to select promising moves.

Importance

- Reduces search time.
- Helps solve complex problems quickly.

- Supports expert systems.

3. Procedural Knowledge

Definition

Procedural Knowledge describes:

How to perform a task.

It is also called **Imperative Knowledge**.

Contains

- Procedures
- Rules
- Strategies
- Algorithms
- Plans

Examples

- How to drive a car.
- How to cook rice.
- How to install software.

AI Example

A robot's instructions for picking and placing an object.

Importance

Provides step-by-step guidance for performing tasks.

4. Declarative Knowledge

Definition

Declarative Knowledge provides information about:

- Facts
- Objects
- Concepts

- Properties

It answers:

"What is it?"

It is also called **Descriptive Knowledge**.

Characteristics

- Easier to represent.
- Expressed through statements.
- Describes facts and attributes.

Examples

- Delhi is the capital of India.
- A triangle has three sides.
- A car has four wheels.

AI Example

Database facts stored in an expert system.

Importance

Provides descriptive information about the world.

Declarative Knowledge	**Procedural Knowledge**
Describes facts	Describes procedures
Answers "What?"	Answers "How?"
Static information	Action-oriented information
Example: Earth is a planet	Example: How to launch a satellite

5. Structural Knowledge

Definition

Structural Knowledge describes relationships among concepts and objects.

It explains:

- How concepts are connected.
- Organization of knowledge.
- Dependencies between concepts.

Relationship Types

a) Kind-of Relationship

Shows category hierarchy.

Example:

Animal → Dog

Dog is a kind of Animal.

b) Part-of Relationship

Shows component relationships.

Example:

Car → Engine

Engine is part of a Car.

c) Grouping Relationship

Shows membership within a collection.

Example:

Students belong to a classroom.

Importance

Structural knowledge helps in:

- Understanding relationships.
- Organizing knowledge.
- Solving complex problems efficiently.

Types of Knowledge Representation (Easy Explanation)

Knowledge Representation (KR) is a way of storing knowledge in a computer so that it can understand, reason, and make decisions.

1. Simple Relational Knowledge

This method stores facts in the form of **tables (relations)**, similar to a database.

Example

Student ID	Name	Course
101	Diya	BCA
102	Ravi	BSc

Here, each row represents an object (student), and columns represent attributes.

Advantages

- Easy to store and retrieve data.
- Similar to database tables.

Disadvantages

- Limited reasoning capability.
- Difficult to derive new knowledge automatically.

2. Inheritable Knowledge

In this method, knowledge is organized in a **hierarchy of classes and subclasses**.

Example

Student
|
|-- Undergraduate Student
| |
| |-- Zinya
|
|-- Postgraduate Student
|
|-- Vidhi

Explanation

- **Student** is a general class.

- **Undergraduate Student** and **Postgraduate Student** are specialized classes.
- **Zinya** and **Vidhi** are instances (individual objects).

This relationship is called an **IS-A relationship**.

- Zinya **IS-A** Undergraduate Student.
- Undergraduate Student **IS-A** Student.

Advantages

- Reduces repetition of information.
- Properties can be inherited from parent classes.

Example

If all students have an ID card, then:

- Undergraduate students automatically have an ID card.
- Postgraduate students automatically have an ID card.

3. Inferential Knowledge

This method represents knowledge using **logic** and allows the system to derive new facts.

Given Statements

Statement 1: Diya is a student.

Student(Diya)

Statement 2: All students are bright.

∀x [Student(x) → Bright(x)]

Inference

Since:

- Diya is a student.
- All students are bright.

Therefore,

Bright(Diya)

Explanation

The system did not store "Diya is bright" directly. It **inferred** (derived) this fact using logical reasoning.

Advantages

- Supports intelligent decision-making.
- Can generate new knowledge from existing facts.

Disadvantages

- More complex to implement.
- Requires logical rules.

4.5 Techniques of Knowledge Representation in AI – Notes

Introduction

Knowledge Representation (KR) is the method of storing knowledge in a form that a computer can understand and use for reasoning, problem-solving, and decision-making.

Techniques of Knowledge Representation

1. Logical Representation
2. Semantic Network Representation
3. Frame Representation
4. Production Rules
5. Propositional Logic

4.5.1 Logical Representation

Definition

Logical Representation uses a formal language with predefined rules to represent knowledge without ambiguity. It helps in drawing conclusions through logical reasoning.

Components

Syntax

- Defines the rules for constructing valid logical statements.

Semantics

- Defines the meaning and interpretation of logical statements.

Types of Logical Representation

1. Propositional Logic

- Uses propositions that are either True or False.
- Example:
    - P: It is raining.
    - Q: The road is wet.

2. First Order Logic (FOL)

- More expressive than propositional logic.
- Uses objects, predicates, and quantifiers.
- Example:
    - Student(Diya)
    - ∀x [Student(x) → Bright(x)]

Advantages

- Precise and unambiguous.
- Supports logical reasoning.
- Easy to verify conclusions.

Disadvantages

- Difficult to represent complex real-world knowledge.
- Computationally expensive.
- Less natural for humans.

4.5.2 Semantic Network Representation

Definition

A Semantic Network represents knowledge using a graphical structure consisting of nodes and links.

Components

Nodes

Represent objects, concepts, or entities.

Arcs (Links)

Represent relationships between objects.

Types of Relations

IS-A Relation

Represents inheritance.

Example:

- Parrot IS-A Bird
- Bird IS-A Animal

Instance Relation

Represents membership.

Example:

- Mittu is an instance of Parrot.

Example

Animal
|
IS-A
|
Bird
|
IS-A
|
Parrot

Advantages

- Easy to understand and visualize.

- Supports inheritance.
- Natural representation of knowledge.

Disadvantages

- Difficult to manage large networks.
- Limited reasoning capabilities.
- Relationships may become complex.

4.5.3 Frame Representation

Definition

A Frame is a record-like structure used to represent knowledge about an object or event through attributes and values.

Components

Slot

An attribute of an object.

Facet

The value associated with a slot.

Example: Frame for a Book

Slot	Value
Title	AI Fundamentals
Author	John Smith
Publisher	ABC Publications
Price	₹500
Pages	350

Characteristics

- Derived from semantic networks.
- Similar to classes and objects in object-oriented programming.

- Frames can be interconnected.

Advantages

- Structured representation.
- Easy inheritance of properties.
- Efficient storage of object information.

Disadvantages

- Difficult to handle complex reasoning.
- Maintenance becomes difficult for large systems.

4.5.4 Production Rules

Definition

Production Rules represent knowledge in the form of IF–THEN statements.

General Form

IF Condition
THEN Action

Examples

Example 1

IF Hungry
THEN Eat

Example 2

IF On Bus AND Bought Ticket AND Vacant Seat
THEN Sit Down

Components of a Production Rule System

1. Set of Production Rules

Collection of IF–THEN rules.

2. Working Memory

Stores the current state of the problem.

3. Recognize–Act Cycle

- Recognize applicable rules.
- Select a rule.
- Execute the corresponding action.

Conflict Resolution

When multiple rules are applicable simultaneously, the system selects the most appropriate rule. This situation is called a Conflict Set.

Advantages

- Simple and easy to understand.
- Modular structure.
- Widely used in expert systems.

Disadvantages

- Large rule sets may be required.
- Conflict resolution can become complex.

4.5.5 Propositional Logic (PL)

Definition

Propositional Logic is the simplest form of logic in which knowledge is represented using propositions.

Proposition

A declarative statement that is either True or False, but not both.

Examples

True Proposition

2 + 2 = 4

False Proposition

The Sun rises in the west.

Important Concepts

Atomic Proposition

A simple proposition consisting of a single statement.

Examples:

- 2 + 2 = 4
- The Sun rises in the east

Compound Proposition

Formed by combining atomic propositions using logical operators.

Examples:

- The Sun rises in the east AND sets in the west.
- It is raining OR it is cloudy.

Logical Operators

Symbol	Meaning
¬P	NOT P
P ∧ Q	P AND Q
P ∨ Q	P OR Q
P → Q	IF P THEN Q
P ↔ Q	IF AND ONLY IF

Tautology

A proposition that is always true.

Example:

P ∨ ¬P

Contradiction

A proposition that is always false.

Example:

P ∧ ¬P

Satisfiable Proposition

A proposition that is true for at least one interpretation.

Example:

P ∨ Q

Contingent Proposition

A proposition that is neither always true nor always false.

Example:

I am not well, so I am taking leave today.

Statements That Are Not Propositions

Questions

- Which course are you opting for?

Commands

- Read the newspaper.

Opinions

- I think it is fiction.

These statements cannot be assigned definite True or False values.

4.6 Syntax of Propositional Logic

Definition

Syntax specifies the rules for constructing valid propositional logic expressions.

Types of Propositions

1. Atomic Proposition

A single indivisible statement.

Examples:

- 2 + 2 = 4
- The Sun rises in the west

2. Compound Proposition

A combination of atomic propositions connected using logical operators.

Examples:

- The Sun rises in the east AND sets in the west.
- Ruskin Bond authored children's books AND received the Sahitya Akademi Award.

Reasoning and Learning in Artificial Intelligence

5.1 Reasoning

Reasoning is the process of analyzing available information, comparing it with existing knowledge, and drawing conclusions.

It helps in:

- Decision making
- Problem solving
- Prediction
- Evaluation of actions

Example

Virtual assistants such as Siri use reasoning to answer questions, provide directions, and make recommendations.

5.1.1 Features of a Reasoning System

A good reasoning system has the following characteristics:

1. Self-Contained System

- Works independently.
- Does not require constant modification by programmers.

2. Solves Different Problems

- Can learn through examples, instructions, or trial-and-error.
- Can solve new and unfamiliar problems.

3. Improves with Experience

- Performance improves after a limited number of trials.

4. Learns Quickly

- Can solve new problems using only a small number of examples.

5.1.2 Why Focus on Reasoning Systems?

Reasoning systems help machines:

- Analyze information.
- Solve problems.
- Make decisions.
- Draw conclusions.

Difference Between Prediction, Inference and Reasoning

Prediction	Inference	Reasoning
Uses pattern matching	Generates new information	Goal-oriented process
Maps inputs to outputs	Automatic process	Produces desired conclusions
Example: Weather forecast	Deriving facts	Decision making

5.1.3 Methods of Reasoning

The major reasoning methods are:

1. Deductive Reasoning
2. Inductive Reasoning
3. Common Sense Reasoning
4. Monotonic Reasoning
5. Non-Monotonic Reasoning
6. Abductive Reasoning

1. Deductive Reasoning

Definition

Deductive reasoning uses general facts or rules to reach specific conclusions.

Approach

Top-Down Approach

General Rule → Specific Conclusion

Example

Premise 1: All humans breathe.

Premise 2: Naina is a human.

Conclusion: Naina breathes.

Characteristics

- Based on facts and rules.
- Produces logically certain conclusions.
- Conclusion is true if premises are true.

Advantages

- Highly reliable.
- Gives accurate conclusions.

Disadvantages

- Requires true facts.
- Difficult to apply in uncertain situations.

2. Inductive Reasoning

Definition

Inductive reasoning uses specific observations to form general conclusions.

Approach

Bottom-Up Approach

Specific Facts → General Rule

Example

- All students scoring 100% are from State X.
- Therefore, students from State X are highly talented.

Characteristics

- Generates hypotheses.
- Conclusion may not always be true.
- Exploratory in nature.

Advantages

- Works with limited data.
- Fast and flexible.

Disadvantages

- Conclusions may be incorrect.
- Less certain than deductive reasoning.

Difference Between Deductive and Inductive Reasoning

Deductive Reasoning	**Inductive Reasoning**
Top-down approach	Bottom-up approach
General to specific	Specific to general
Produces certain conclusions	Produces probable conclusions
Requires facts	Requires evidence

3. Common Sense Reasoning

Definition

Reasoning based on everyday experiences and practical knowledge.

Examples

- Wearing a mask during a lockdown to avoid fines.
- Looking both ways before crossing a road.

Characteristics

- Informal reasoning.
- Based on experiences.

- Uses good judgment.
- Uses heuristic knowledge.

Applications

- Computer Vision
- Robotics
- Intelligent Assistants

Advantages

- Mimics human thinking.
- Useful in real-world situations.

Disadvantages

- Not always logically accurate.
- Depends on personal experiences.

4. Monotonic Reasoning

Definition

Once a conclusion is derived, it never changes even if new information is added.

Examples

- The Harappan Civilization existed in the Indus Valley.
- Kashmir is famous for natural beauty.

Characteristics

- Conclusions remain fixed.
- New facts do not affect old conclusions.
- Used in logic-based systems.

Advantages

1. Proofs remain valid.
2. Derived facts remain valid forever.

Disadvantages

1. Cannot handle real-world uncertainty.
2. Cannot revise conclusions.
3. Cannot represent hypothetical knowledge.

5. Non-Monotonic Reasoning

Definition

A conclusion can change when new information becomes available.

Example

Knowledge Base:

- Birds fly.
- Tweety is a bird.

Conclusion:

- Tweety flies.

Later information:

- Tweety is a penguin.

New Conclusion:

- Tweety does not fly.

Characteristics

- Handles incomplete information.
- Supports changing conclusions.
- Represents real-world situations.

Advantages

1. Suitable for real-world applications.
2. Supports assumptions and probabilities.

Disadvantages

1. Old conclusions may become invalid.
2. Not suitable for theorem proving.

6. Abductive Reasoning

Definition

Abductive reasoning starts with observations and finds the most likely explanation.

Example

A doctor diagnoses a disease using current test reports.

Characteristics

- Uses present observations.
- Finds the most probable explanation.
- Useful for decision-making and troubleshooting.

Advantages

- Useful in diagnosis.
- Helps make decisions under uncertainty.

Disadvantages

- Conclusions may not always be correct.
- Can justify incorrect beliefs.

5.2 Probabilistic Reasoning in AI

Probabilistic Reasoning is a technique that combines **probability theory** with **logic** to handle uncertainty in Artificial Intelligence.

Why is Probabilistic Reasoning Needed?

In real-world situations:

- Information may be incomplete.
- Facts may be uncertain.
- Outcomes may be unpredictable.

- Errors may occur during observations.

Example

"It may rain today."

We cannot say with complete certainty whether it will rain or not. Therefore, probability is used to express such uncertainty.

5.2.1 Uncertainty

Definition

Uncertainty occurs when we are not completely sure whether a statement or event is true or false.

Causes of Uncertainty

1. Unreliable information sources
2. Experimental errors
3. Equipment faults
4. Temperature variations
5. Climate changes

Example

A weather forecast predicts a 70% chance of rain. Rain may or may not occur, creating uncertainty.

5.2.2 Probability

Definition

Probability is the measure of the likelihood that an event will occur.

Range of Probability

$0 \leq P(A) \leq 1$

Where:

- $P(A) = 0 \rightarrow$ Event will never occur (Impossible Event)

- P(A) = 1 → Event will definitely occur (Certain Event)

Example

- Probability of the sun rising tomorrow = 1
- Probability of rolling a 7 on a six-sided die = 0

Complement Rule

The probability of an event occurring and not occurring adds up to 1.

Formula

Where:

- P(A) = Probability of event A occurring
- P(¬A) = Probability of event A not occurring

Example

If probability of rain is 0.7,

Then:

P(¬Rain) = 1 - 0.7 = 0.3

Important Terms in Probability

1. Event

Definition

An event is a possible outcome of an experiment.

Example

Coin Toss:

- Head
- Tail

Each outcome is called an event.

2. Sample Space

Definition

The set of all possible outcomes of an experiment.

Example

For a coin toss:

S = {Head, Tail}

For rolling a die:

S = {1, 2, 3, 4, 5, 6}

3. Prior Probability

Definition

Probability calculated before observing any new information.

Example

Before checking the weather report:

Probability of rain = 0.4

This is Prior Probability.

4. Posterior Probability

Definition

Probability calculated after receiving new evidence or information.

Example

After seeing dark clouds:

Probability of rain = 0.8

This updated probability is called Posterior Probability.

5. Independent Events

Definition

Events that do not affect each other.

Example

Tossing a coin twice.

The result of the first toss does not affect the second toss.

Formula

$P(A|B) = P(A)$

Easy Example

- First Toss = Head
- Second Toss = Tail

Both events are independent.

6. Dependent Events

Definition

Events whose outcomes depend on previous events.

Example

A bag contains:

- 2 Blue balls
- 3 Red balls

If one ball is removed without replacement, the next draw depends on the first draw.

7. Marginal Probability

Definition

Probability of a single event occurring without considering any other event.

Example

Probability of drawing a red card from a deck:

P(A) = 26/52 = 0.5

Meaning:

There is a 50% chance of drawing a red card.

8. Joint Probability

Definition

Probability of two events occurring together.

Formula

P(A ∩ B)

Example

Probability of drawing a card that is:

- Red
- Number 8

P(A ∩ B) = 2/52 = 1/26

9. Conditional Probability

Definition

Conditional probability is the probability of an event occurring when another event has already occurred.

Formula

Where:

- P(A∩B) = Joint Probability
- P(B) = Probability of Event B

Easy Example

Suppose:

- 30 students like both Cricket and Football.
- 65 students like Football.

Find the probability that a student likes Cricket given that the student likes Football.

$P(A \cap B)=30/100=0.30$
$P(B)=65/100=0.65$

$P(A|B)=0.30/0.65$
≈ 0.46

Therefore,

There is a 46% probability that a student likes Cricket given that the student likes Football.

Solved Example

Question

70% students want to learn R language.

40% students want to learn Python.

Find the percentage of students who like both R and Python.

Solution

Let:

- A = Students liking Python
- B = Students liking R

Given:

$P(A|B)=0.40$
$P(B)=0.70$

Using:

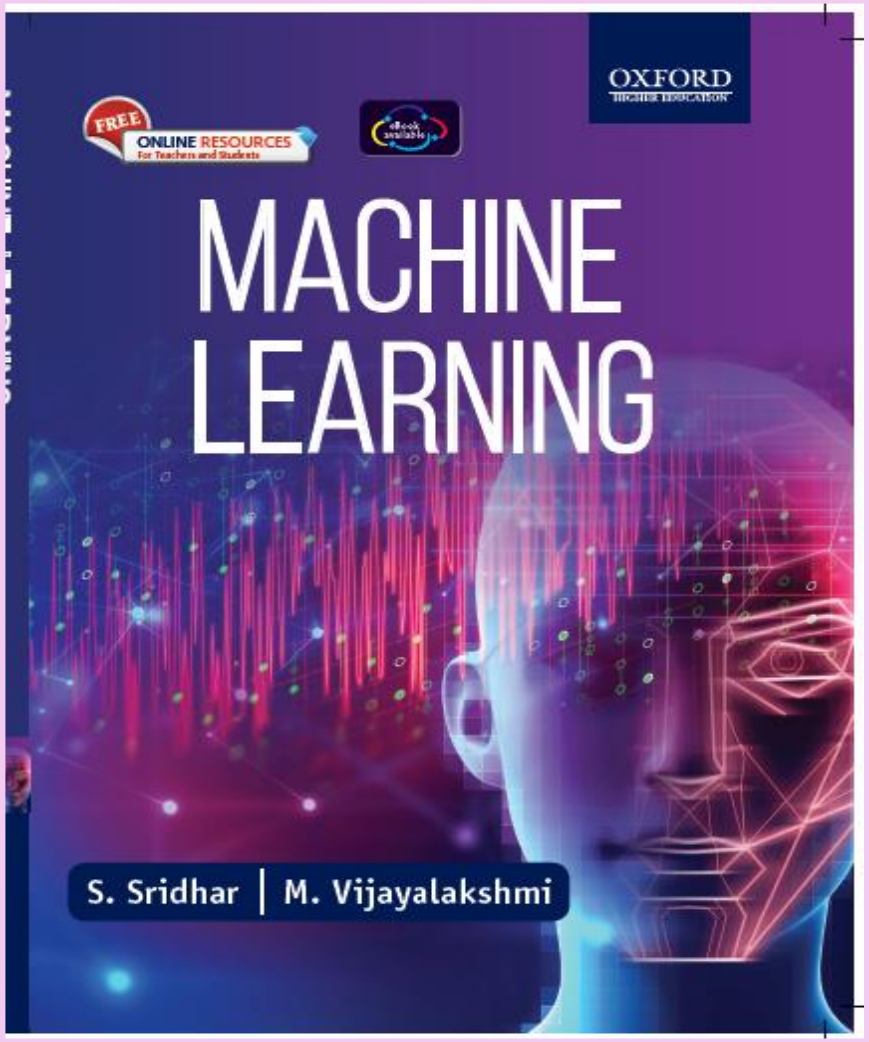
$P(A \cap B)=P(A|B) \times P(B)$

$P(A \cap B)=0.40 \times 0.70$
$=0.28$

Answer

$0.28 \times 100 = 28\%$

Therefore, 28% of students like both R and Python.



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Machine Learning

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Chapter 1

Introduction to Machine Learning

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Need for Machine Learning?

- BUSINESS ORGANIZATIONS HAVE NUMEROUS DATA
- NEED TO ANALYZE DATA FOR TAKING DECISIONS

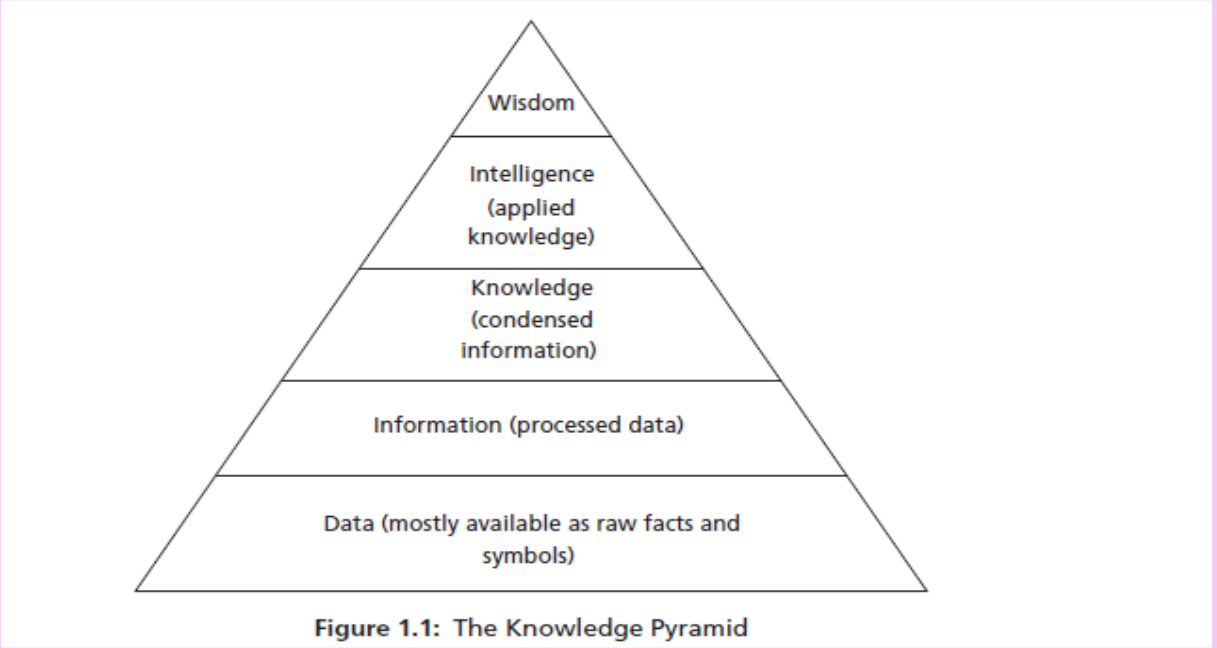


Figure 1.1: The Knowledge Pyramid

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Popularity for Machine Learning

- HIGH VOLUME PF DATA
- REDUCED COST OF STORAGE
- AVAILABILITY OF COMPLEX ALGORITHMS AND TOOLS

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What is Machine Learning?

- "MACHINE LEARNING IS A FIELD OF STUDY THAT GIVES THE COMPUTERS TO LEARN WITHOUT BEING EXPLICITLY PROGRAMMED"

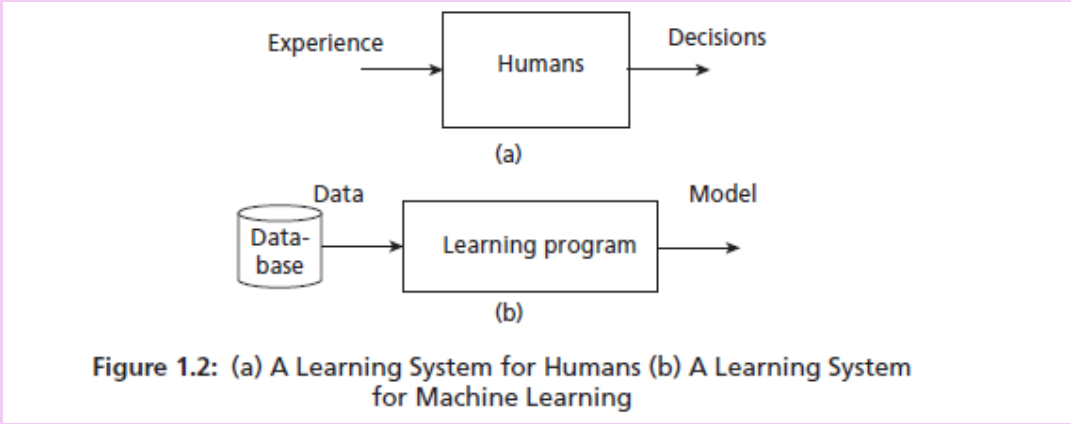


Figure 1.2: (a) A Learning System for Humans (b) A Learning System for Machine Learning

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What is a Model?

A MODEL CAN BE ANY ONE OF THE FOLLOWING –

- MATHEMATICAL EQUATION
- RELATIONAL DIAGRAMS LIKE GRAPHS/TREES
- LOGICAL IF/ELSE RULES
- GROUPINGS CALLED CLUSTERS

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Another View of Machine Learning

- TOM MITCHELL DEFINITION OF MACHINE LEARNING

Another pioneer of AI, Tom Mitchell's definition of machine learning states that, "*A computer program is said to learn from experience E, with respect to task T and some performance measure P, if its performance on T measured by P improves with experience E.*" The important components of this definition are experience *E*, task *T*, and performance measure *P*.

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Machine Learning and AI

- RELATION BETWEEN MACHINE LEARNING AND AI

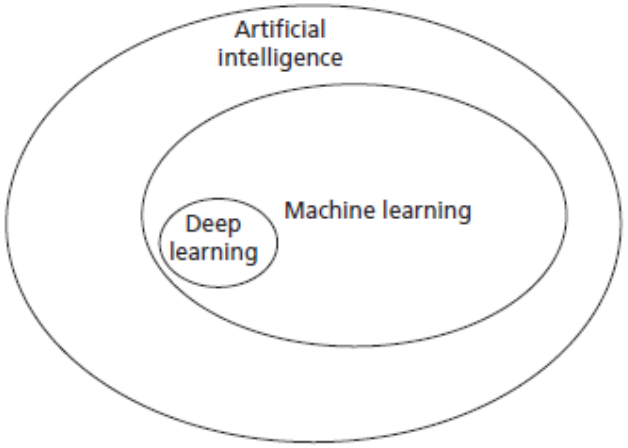


Figure 1.3: Relationship of AI with Machine Learning

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Machine Learning and Data Science

- DATA SCIENCE IS AN "UMBRELLA TERM" COVERING FROM DATA COLLECTION TO DATA ANALYSIS.

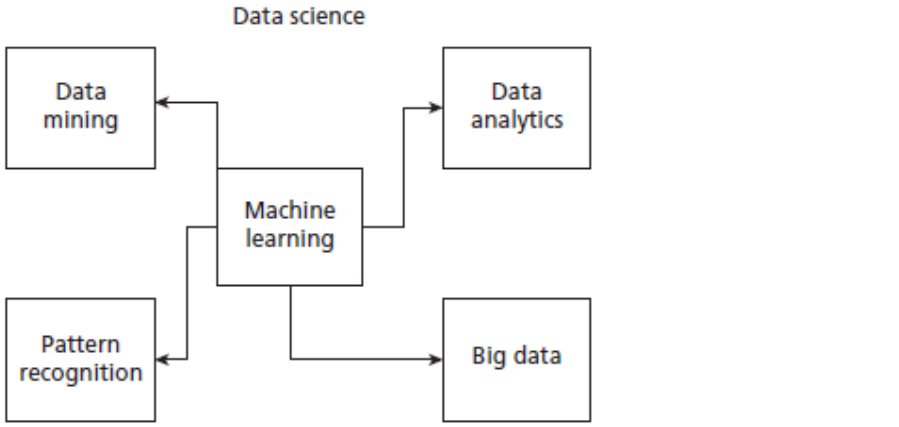


Figure 1.4: Relationship of Machine Learning with Other Major Fields

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Machine Learning and Data Science

- CHARACTERISTICS OF BIG DATA

Big Data Data science concerns about collection of data. Big data is a field of data science that deals with data's following characteristics:

1. Volume: Huge amount of data is generated by big companies like Facebook, Twitter, YouTube.
2. Variety: Data is available in variety of forms like images, videos, and in different formats.
3. Velocity: It refers to the speed at which the data is generated and processed.

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Machine Learning and Data Science

- DATA SCIENCE AND DATA MINING

Data Mining Data mining's original genesis is in the business. Like while mining the earth one gets into precious resources, it is often believed that unearthing of the data produces hidden information that otherwise would have eluded the attention of the management. Nowadays, many consider that data mining and machine learning are same. There is no difference between these fields except that data mining aims to extract the hidden patterns that are present in the data, whereas, machine learning aims to use it for prediction.

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Machine Learning and Data Science

- DATA SCIENCE AND DATA ANALYTICS / PATTERN RECOGNITION

Data Analytics Another branch of data science is data analytics. It aims to extract useful knowledge from crude data. There are different types of analytics. Predictive data analytics is used for making predictions. Machine learning is closely related to this branch of analytics and shares almost all algorithms.

Pattern Recognition It is an engineering field. It uses machine learning algorithms to extract the features for pattern analysis and pattern classification. One can view pattern recognition as a specific application of machine learning.

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Machine Learning and Statistics

ROLE OF STATISTICS

Statistics is a branch of mathematics that has a solid theoretical foundation regarding statistical learning. Like machine learning (ML), it can learn from data. But the difference between statistics and ML is that statistical methods look for regularity in data called patterns. Initially, statistics sets a hypothesis and performs experiments to verify and validate the hypothesis in order to find relationships among data.

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Machine Learning Types

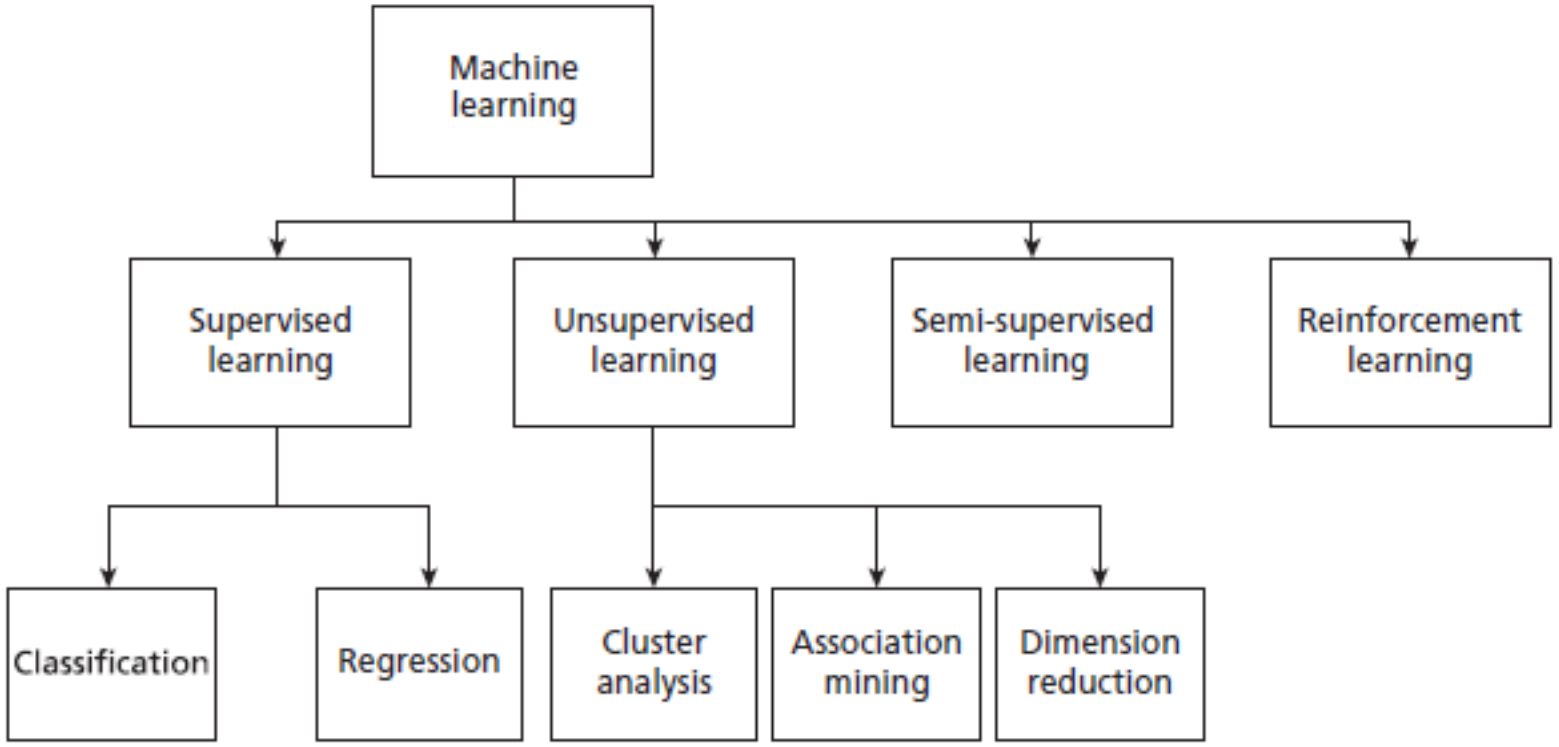


Figure 1.5: Types of Machine Learning

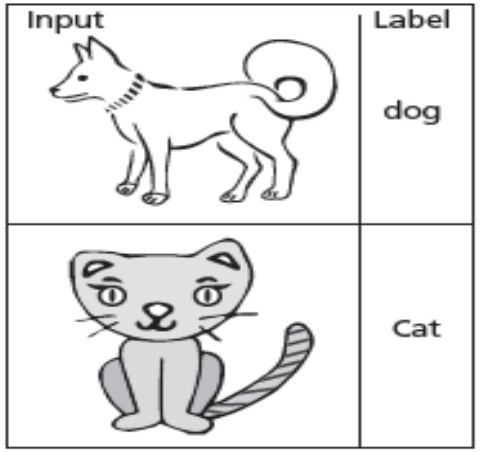
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Labelled Data

Table 1.1: Iris Flower Dataset

S.No.	Length of Petal	Width of Petal	Length of Sepal	Width of Sepal	Class
1.	5.5	4.2	1.4	0.2	Setosa
2.	7	3.2	4.7	1.4	Versicolor
3.	7.3	2.9	6.3	1.8	Virginica

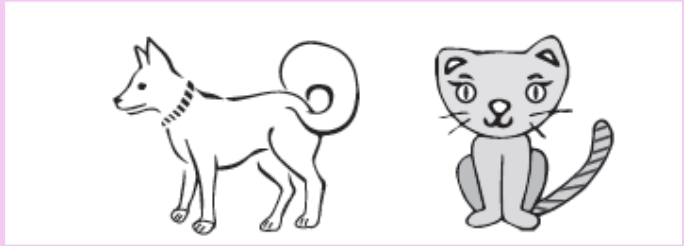
A dataset need not be always numbers. It can be images or video frames. Deep neural networks can handle images with labels. In the following Figure 1.6, the deep neural network takes images of dogs and cats with labels for classification.



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Unlabelled Data

DATA THAT IS NOT ASSOCIATED WITH LABELS IS CALLED UNLABELLED DATA.



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Supervised Learning

CLASSSIFICATION

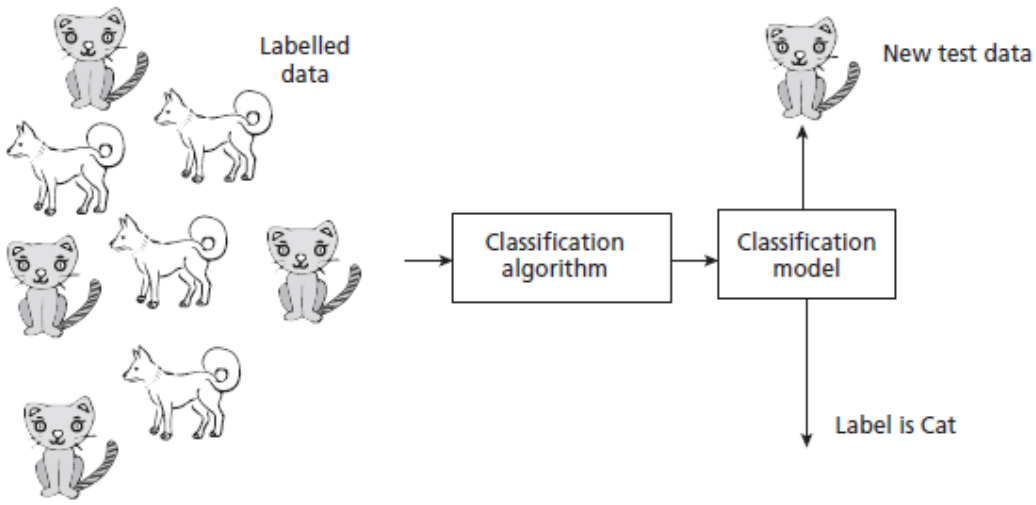


Figure 1.7: An Example Classification System

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Supervised Learning

KEY ALGORITHMS

Some of the key algorithms of classification are:

- Decision Tree
- Random Forest
- Support Vector Machines
- Naïve Bayes
- Artificial Neural Network and Deep Learning networks like CNN

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Supervised Learning

REGRESSION ALGORITHMS

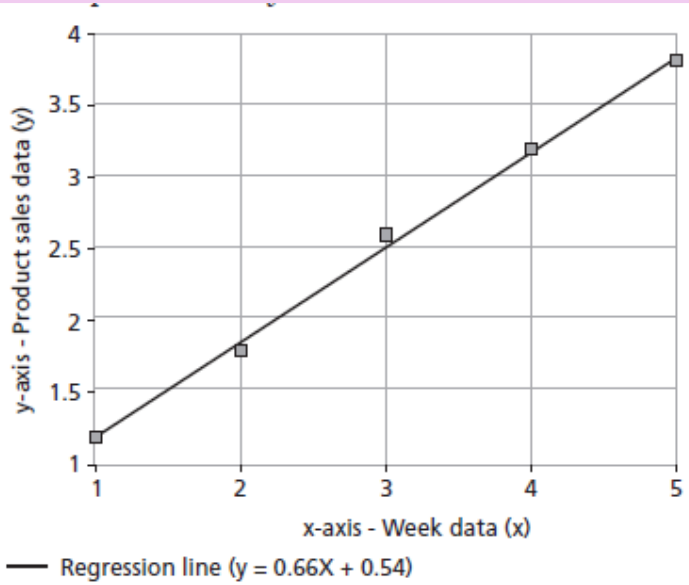


Figure 1.8: A Regression Model of the Form $y = ax + b$

product sales = 0.66 × Week + 0.54.

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Unsupervised Learning

CLUSTERING IS A GROUPING PROCESS.

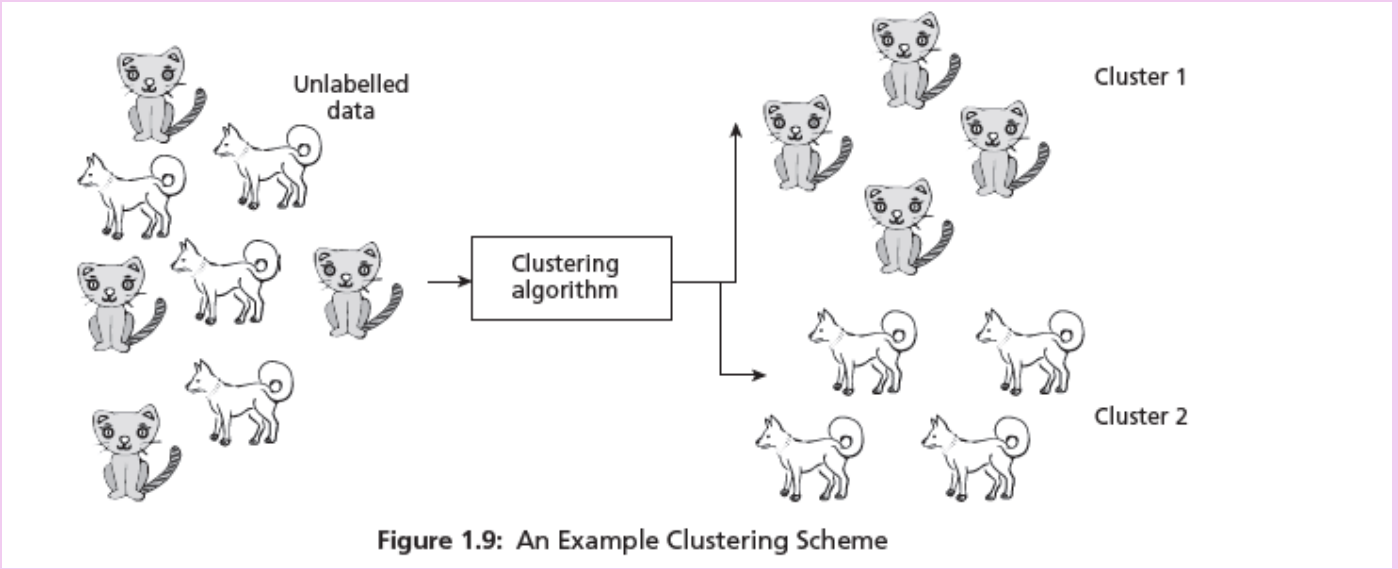


Figure 1.9: An Example Clustering Scheme

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Unsupervised Learning

KEY ALGORITHMS OF UNSUPERVISED LEARNING

- k-means algorithm
- Hierarchical algorithms

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Key Differences

S.No.	Supervised Learning	Unsupervised Learning
1.	There is a supervisor component	No supervisor component
2.	Uses Labelled data	Uses Unlabelled data
3.	Assigns categories or labels	Performs grouping process such that similar objects will be in one cluster

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Semi-supervised Learning

There are circumstances where the dataset has a huge collection of unlabelled data and some labelled data. Labelling is a costly process and difficult to perform by the humans. Semi-supervised algorithms use unlabelled data by assigning a pseudo-label. Then, the labelled and pseudo-labelled dataset can be combined.

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Reinforcement Learning

Reinforcement learning mimics human beings. Like human beings use ears and eyes to perceive the world and take actions, reinforcement learning allows the agent to interact with the environment to get rewards. The agent can be human, animal, robot, or any independent program. The rewards enable the agent to gain experience. The agent aims to maximize the reward.

The reward can be positive or negative (Punishment). When the rewards are more, the behavior gets reinforced and learning becomes possible.

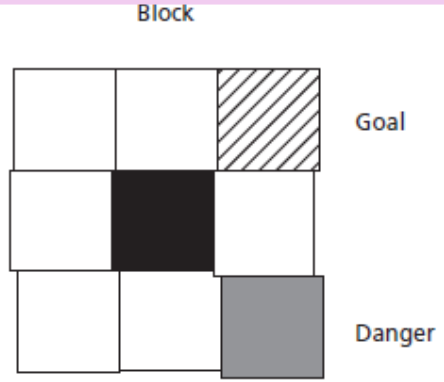


Figure 1.10: A Grid game

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Challenges of Machine Learning

1. ILL-POSED PROBLEMS – PROBLEMS WHOSE SPECIFICATIONS ARE NOT CLEAR
2. HUGE DATA
3. HUGE COMPUTATION POWER
4. COMPLEXITY OF ALGORITHMS
5. BIAS-VARIANCE

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Machine Learning Process

MACHINE LEARNING/DATA MINING PROCESS

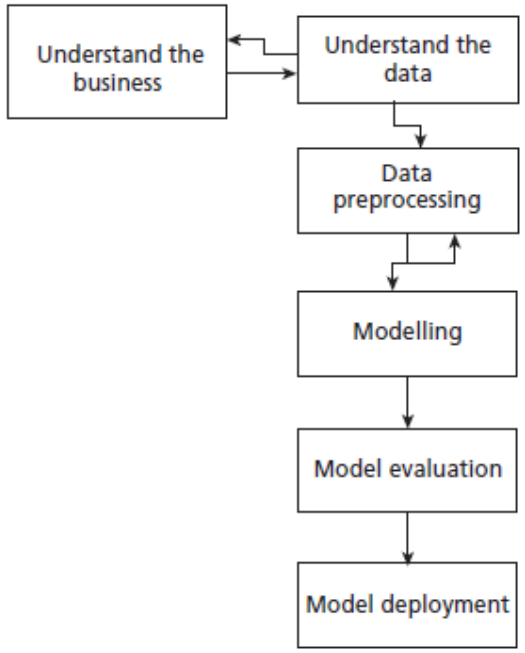


Figure 1.11: A Machine Learning/Data Mining Process

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Machine Learning Applications

MACHINE LEARNING MAJOR APPLICATIONS

1. Sentiment analysis – This is an application of natural language processing (NLP) where the words of documents are converted to sentiments like happy, sad, and angry which are captured by emoticons effectively. For movie reviews or product reviews, five stars or one star are automatically attached using sentiment analysis programs.
2. Recommendation systems – These are systems that make personalized purchases possible. For example, Amazon recommends users to find related books or books bought by people who have the same taste like you, and Netflix suggests shows or related movies of your taste. The recommendation systems are based on machine learning.
3. Voice assistants – Products like Amazon Alexa, Microsoft Cortana, Apple Siri, and Google Assistant are all examples of voice assistants. They take speech commands and perform tasks. These chatbots are the result of machine learning technologies.
4. Technologies like Google Maps and those used by Uber are all examples of machine learning which offer to locate and navigate shortest paths to reduce time.

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Machine Learning Applications

S.No.	Problem Domain	Applications
1.	Business	Predicting the bankruptcy of a business firm
2.	Banking	Prediction of bank loan defaulters and detecting credit card frauds
3.	Image Processing	Image search engines, object identification, image classification, and generating synthetic images
4.	Audio/Voice	Chatbots like Alexa, Microsoft Cortana. Developing chatbots for customer support, speech to text, and text to voice
5.	Telecommuni-cation	Trend analysis and identification of bogus calls, fraudulent calls and its callers, churn analysis
6.	Marketing	Retail sales analysis, market basket analysis, product performance analysis, market segmentation analysis, and study of travel patterns of customers for marketing tours
7.	Games	Game programs for Chess, GO, and Atari video games
8.	Natural Language Translation	Google Translate, Text summarization, and sentiment analysis
9.	Web Analysis and Services	Identification of access patterns, detection of e-mail spams, viruses, personalized web services, search engines like Google, detection of promotion of user websites, and finding loyalty of users after web page layout modification
10.	Medicine	Prediction of diseases, given disease symptoms as cancer or diabetes. Prediction of effectiveness of the treatment using patient history and Chatbots to interact with patients like IBM Watson uses machine learning technologies.
11.	Multimedia and Security	Face recognition/identification, biometric projects like identification of a person from a large image or video database, and applications involving multimedia retrieval
12.	Scientific Domain	Discovery of new galaxies, identification of groups of houses based on house type/geographical location, identification of earthquake epicenters, and identification of similar land use

Summary

1. Machine learning can enable top management of an organization to extract the knowledge from the data stored in various archives to facilitate decision making.
2. Machine learning is an important subbranch of Artificial Intelligence (AI).
3. A model is an explicit description of patterns within the data.
4. A model can be a formula, procedure or representation that can generate data decisions.
5. Humans predict by remembering the past, then formulate the strategy and make a prediction. In the same manner, the computers can predict by following the process.
6. Machine learning is an important branch of AI. AI is a much broader subject. The aim of AI is to develop intelligent agents. An agent can be a robot, humans, or other autonomous systems.

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Summary

7. Deep learning is a branch of machine learning. The difference between machine learning and deep learning is that models are constructed using neural network technology in deep learning. Neural networks are models constructed based on the human neuron models.
8. Data science deals with gathering of data for analysis. It is a broad field that includes other fields.
9. Data analytics aims to extract useful knowledge from crude data. There are many types of analytics. Predictive data analytics is an area that is dedicated for making predictions. Machine learning is closely related to this branch of analytics and shares almost all algorithms.
10. One can say thus there are two types of data – labelled data and unlabelled data. The data with a label is called labelled data and those without a label are called unlabelled data.
11. Supervised algorithms use labelled dataset. As the name suggests, there is a supervisor or teacher component in supervised learning. A supervisor provides the labelled data so that the model is constructed and gives test data for checking the model.
12. Classification is a supervised learning method. The input attributes of the classification algorithms are called independent variables. The target attribute is called label or dependent variable. The relationship between the input and target variables is represented in the form of a structure which is called a classification model.
13. Cluster analysis is an example of unsupervised learning. It aims to assemble objects into disjoint clusters or groups.
14. Semi-supervised algorithms assign a pseudo-label for unlabelled data.
15. Reinforcement learning allows the agent to interact with the environment to get rewards. The agent can be human, animal, robot, or any independent program. The rewards enable the agent to gain experience.
16. The emerging process model for the data mining solutions for business organizations is CRISP-DM. This model stands for Cross Industry Standard Process – Data Mining.
17. Machine Learning technologies are used widely now in different domains.

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Design of Learning System

Definition

A **learning system** is a system that improves its performance automatically by learning from experience.

It learns from past data and becomes better at performing a task over time.

A learning system is defined using three components:

- **Task (T)**
- **Experience (E)**
- **Performance (P)**

A system learns if its performance improves with experience.

Design of a Learning System

1. Choosing the Training Experience (E)

This step decides:

- What type of data the system will learn from

- Types of Experience:

Direct Experience

- System interacts with environment
- Example: Chess-playing program learns by playing games

Indirect Experience

- Uses stored dataset
- Example: Student dataset (CGPA, skills, result)

Example:

A placement prediction system:

- Input data → CGPA, communication skills

- Output → Selected / Not Selected

This dataset is the **training experience**

2. Determining the Target Function (f)

The **target function** defines:

What the system is supposed to learn
It maps:
Input → Output

Representation:

$$f : X \to Y$$

Example:

f(CGPA, Skills) → Job Selection

The system tries to learn this function

3. Representation of Target Function

The system must decide:

How to represent the function (model)

Common Representations:

1. Mathematical Model

Example:

Salary = 2 × CGPA + 5

2. Decision Tree

CGPA > 7?

Yes → Selected

No → Not Selected

3. Logical Rules

IF CGPA > 7 AND Skills = Good → Selected

4. Neural Networks

Used for complex patterns

Choosing the right representation is very important:

- Simple problems → Linear models
- Complex problems → Trees or Neural Networks

4. Choosing the Learning Algorithm

The learning algorithm:

Finds the best function (hypothesis) from data

Role of Algorithm:

- Searches through possible solutions
- Selects the best one
- Minimizes error

The learning algorithm minimizes the error between the predicted output and the Actual output.

Examples of Algorithms:

- Linear Regression
- Decision Tree (ID3)
- K-Nearest Neighbors
- Neural Networks

A **learning system** is a system that improves its performance automatically by learning from experience.

Designing a learning system involves **four major steps**:

1. Choosing Training Experience

- Training data represents past experience.
- It may be:
    - Direct experience (e.g., playing chess moves)
    - Indirect experience (recorded data)

Example:

- Student dataset with CGPA, skills, and placement status

2. Determining the Target Function

- The **target function (f)** defines what the system must learn.
- It maps input → output

Example:

f(CGPA, Skills) → Job Selection

3. Representation of Target Function

- The function can be represented in different forms:
  - Mathematical equation
  - Decision tree
  - Logical rules

Example:

IF CGPA > 7 AND Skills = Good → Selected

4. Choosing Learning Algorithm

- Algorithm selects the **best hypothesis**
- Goal: minimize error

Example:

- Linear Regression
- Decision Tree
- KNN

Error Minimization

The algorithm tries to reduce error:

$$E = \sum (V_{train}(b) - V(b))^2$$

- E → Total error
- $V_{train}(b)$→ Target (expected/true) value
- V(b)→ Predicted value by the model

Modelling in Machine Learning

What is Modelling?

Modelling = Creating a mathematical representation of a real-world problem

In simple words:

A **model** is something that learns from data and makes predictions.

Example to Understand

Suppose:

Input → CGPA, Communication Skills

Output → Job Selection (Yes/No)

A model will learn a rule like:

IF CGPA > 7 AND Communication = Good → Selected

This rule = **Model**

Why Do We Need Models?

- To **predict future outcomes**
- To **find patterns in data**
- To **automate decisions**

Types of Models:

- Linear models
- Tree models
- Probabilistic models
- Instance-based models

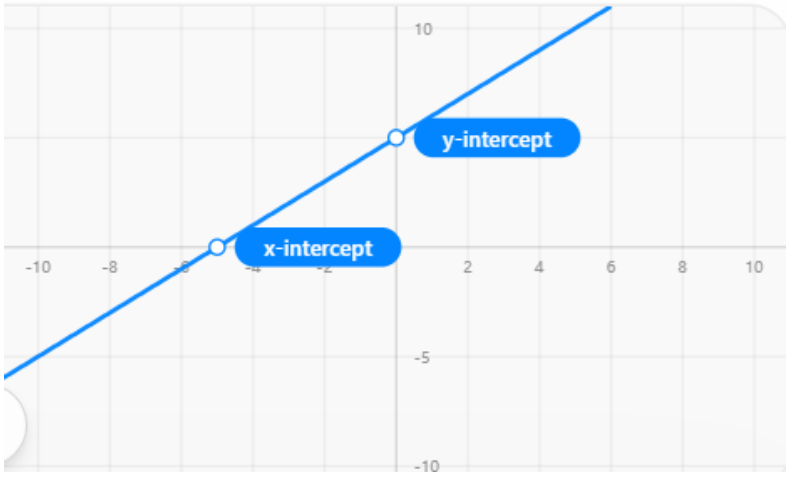
1. Linear Models

Definition

Linear models assume a **linear (straight-line) relationship** between input and output.

Mathematical Form

$$y = mx + b$$



Explanation

- Output changes proportionally with input
- Simple and easy to understand

Example

- Predict salary using CGPA
- If CGPA increases → salary increases

Algorithms

- Linear Regression
- Logistic Regression

Advantages

- Simple
- Fast computation

Disadvantages

- Cannot handle complex relationships

2. Tree Models

Definition

Tree models represent decisions in the form of a **tree structure**.

Structure

- Root node → main decision
- Internal node → condition
- Leaf node → output

Example

CGPA > 7?
YES → Selected
NO → Not Selected

Algorithms

- Decision Tree
- Random Forest

Advantages

- Easy to understand
- Works for classification & regression

Disadvantages

- Can overfit data

3. Probabilistic Models

Definition

These models use **probability theory** to make predictions.

They estimate the **likelihood (chance)** of outcomes.

Example

- Spam detection:
  - Probability email is spam = 0.8 → classify as spam

Algorithms

- Naive Bayes
- Bayesian Networks

Advantages

- Handles uncertainty well
- Works well with small data

Disadvantages

- Assumptions may not always be correct

4. Instance-Based Models

Definition

These models **store training data** and make predictions based on similarity.

Also called **lazy learning**

Key Idea

"Similar inputs → similar outputs"

Example

- New student compared with similar past students
- Prediction based on nearest neighbors

Algorithms

- K-Nearest Neighbors (KNN)

Distance Concept

$$d(x, x_i)$$

Advantages

- Simple
- No training required

Disadvantages

- Slow during prediction
- Requires large memory

Model Type	Idea	Example Algorithm
Linear	Straight-line relationship	Linear Regression
Tree	Decision-based structure	Decision Tree
Probabilistic	Uses probability	Naive Bayes
Instance-based	Uses similarity	KNN

Similarity-Based Learning

Nearest Neighbor Learning

- Store all data
- For a new input → find closest example

Example:
If a student is similar to past selected students → predict "Selected"

Weighted K-Nearest Neighbor (KNN)

Instead of 1 neighbor → take **K neighbors**

"Closer neighbors have more importance"

3. Regression Analysis

What is Regression?

Predict **continuous values**

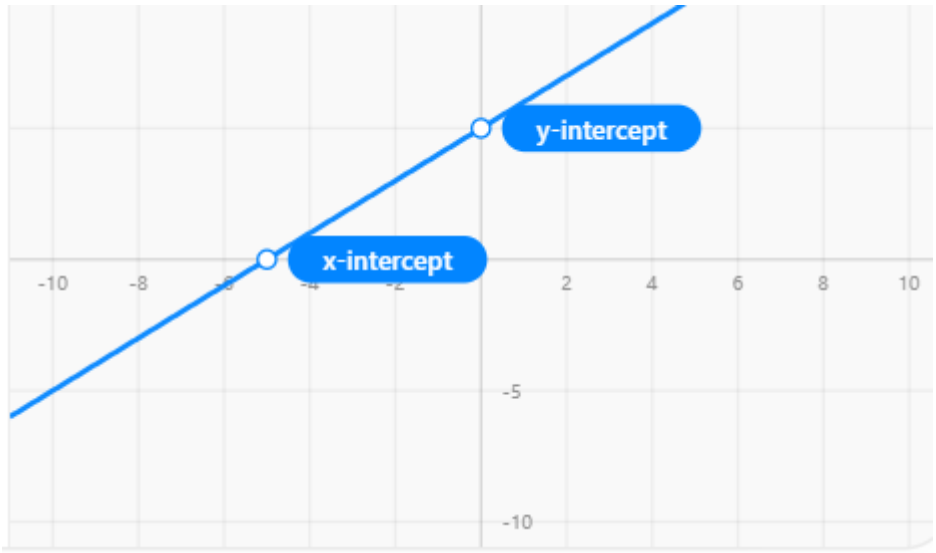
Examples:

- Salary prediction
- House price prediction

Linear Regression

Find a straight line:

$$y = mx + b$$



Where:

- y = output
- x = input
- m = slope
- b = intercept

Example:

CGPA → Salary
Higher CGPA → higher salary

Multiple Linear Regression

More than one input:

$$y = b_0 + b_1x_1 + b_2x_2 + \cdots + b_nx_n$$

Example:

Predict salary using:

- CGPA
- Communication
- Projects

4. Decision Tree Learning

What is Decision Tree?

A tree-like model:

- Each node = decision
- Each branch = outcome
- Leaf = final answer

Example:

CGPA > 7?
YES → Selected
NO → Not Selected

Module 4 :

Basic Learning Theory

3.1 INTRODUCTION TO LEARNING AND ITS TYPES

The process of acquiring knowledge and expertise through study, experience, or being taught is called as learning.

First let us address the question - What sort of tasks can the computers learn? This depends on the nature of problems that computers can solve. There are two kinds of problems - well-posed and ill-posed. Computers can solve only well-posed problems, as these have well-defined specifications and have the following components inherent to it.

1. Class of learning tasks (T)
2. A measure of performance (P)
3. A source of experience (E)

The standard definition of learning proposed by Tom Mitchell is that a program can learn from E for the task T, and P improves with experience E. Let us formalize the concept of learning as follows:

Let Xbe the input and $X_b$be the input space, which is the set of all inputs, and $Y$is the output space, which is the set of all possible outputs, that is, yes/no.

Let Dbe the input dataset with examples,

$$D = \{x_1, y_1\}, \{x_2, y_2\}, \dots, \{x_n, y_n\}$$

for $n$inputs.

Let the unknown target function be:

$$f: X \to Y$$

that maps the input space to output space. The objective of the learning program is to pick a function,

$$g: X \to Y$$

to approximate hypothesis f.

All the possible formulae form a hypothesis space. In short, let Hbe the set of all formulae from which the learning algorithm chooses. The choice is good when the hypothesis $g$replicates $f$for all samples. This is shown in Figure 3.1.

Figure 3.1: Learning Environment

- Training samples
- Candidate formulae
- Hypothesis space H
- Learning algorithms
- Generated hypothesis $g(x)$

- Ideal target hypothesis $f(x)$
- Difference
- Error

It can be observed that training samples and target function are dependent on the given problem. The learning algorithm and hypothesis set are independent of the given problem. Thus, learning model is informally the hypothesis set and learning algorithm. Thus, learning model can be stated as follows:

$$\text{Learning Model} = \text{Hypothesis Set} + \text{Learning Algorithm}$$

Learning Types

There are different types of learning. Some of the different learning methods are as follows:

1. **Learn by memorization** or learn by repetition also called as rote learning is done by memorizing without understanding the logic or concept.
2. **Learn by examples** It is learning from past examples or experiences. The learner finds general rules or patterns from observations.
3. **Learn by being taught** by an expert or a teacher, generally called as passive learning. However, there is a special kind of learning called active learning where the learner can interactively query a teacher/expert to label unlabelled data instances with the desired outputs.
4. **Learning by critical thinking**, also called as deductive learning, deduces new facts or conclusion from related known facts and information.
5. **Self-learning**, also called as reinforcement learning, is a self-directed learning that normally learns from mistakes punishments and rewards.
6. **Learning to solve problems** It is learning by finding methods to achieve a goal. The learner develops steps and strategies to solve problems.
7. **Learning by generalizing explanations**, also called as explanation-based learning (EBL), is another learning method that exploits domain knowledge from experts to improve the accuracy of learned concepts by supervised learning.

3.2 INTRODUCTION TO COMPUTATIONAL LEARNING THEORY

There are many questions that have been raised by mathematicians and logicians over the time taken by computers to learn. Some of the questions are as follows:

1. How can a learning system predict an unseen instance?
2. How do the hypothesis $h$is close to f, when hypothesis $f$itself is unknown?
3. How many samples are required?
4. Can we measure the performance of a learning system?
5. Is the solution obtained local or global?

Computational Learning Theory uses many concepts from diverse areas such as Theoretical Computer Science, Artificial Intelligence and Statistics. The core concept of COLT is the concept of learning

framework. One such important framework is PAC. The learning framework is discussed in a detailed manner in Section 3.7. COLT focuses on supervised learning tasks. Since the complexity of analyzing is difficult, normally, binary classification tasks are considered for analysis.

3.3 DESIGN OF A LEARNING SYSTEM

A system that is built around a learning algorithm is called a learning system. The design of systems focuses on these steps:

1. Choosing a training experience
2. Choosing a target function
3. Representation of a target function
4. Function approximation

Trainingexperience

Training experience means the data or examples used to teach a machine learning system. The system can learn either from direct guidance by a teacher or by practicing and learning from its own experience.

Determine the Target Function

The next step is the determination of a target function. In this step, the type of knowledge that needs to be learnt is determined. In direct experience, a board move is selected and is determined whether it is a good move or not against all other moves. If it is the best move, then it is chosen as:

$$B \rightarrow R$$

where, $B$and $R$are legal moves.

In indirect experience, all legal moves are accepted and a score is generated for each. The move with largest score is then chosen and executed.

Determine the Target Function Representation

The representation of knowledge may be a table, collection of rules or a neural network. The linear combination of these factors can be coined as:

$$V = b + w_1x_1 + w_2x_2 + w_3x_3 + \cdots + w_nx_n$$

where, $x_1, x_2, x_3, \ldots, x_n$represent different board features and $w_1, w_2, w_3, \ldots, w_n$represent weights.

Choosing an Approximation Algorithm for the Target Function

The focus is to choose weights and fit the given training samples effectively. The aim is to reduce the error given as:

$$E = \sum_{i=1}^{n} \left(V_{train}(b_i) - \hat{V}(b_i)\right)^2$$

Here, $b_i$is the sample and $\hat{V}(b_i)$is the predicted hypothesis.

The approximation is carried out as:

- Computing the error as the difference between trained and expected hypothesis. Let error be $error(b_i)$.
- Then, for every board feature x_i, the weights are updated as:

$$w_i = w_i + \mu \times error(b_i) \times x_i$$

Here, $\mu$is the constant that moderates the size of the weight update.
Thus, the learning system has the following components:

- A performance system to allow the game to play against itself.
- A critic system to generate the samples.
- A generalizer system to generate a hypothesis based on samples.
- An Experimenter system to generate a new system based on the currently learnt function. This is sent as input to the performance system.

3.4 INTRODUCTION TO CONCEPT LEARNING

Concept learning is a learning strategy of acquiring abstract knowledge or inferring a general concept or deriving a category from the given training samples. It is a process of abstraction and generalization from the data.

Concept learning requires three things:

1. Input – Training dataset which is a set of training instances, each labelled with the name of a concept or category to which it belongs. Use this past experience to train and build the model.
2. Output – Target concept or Target function f. It is a mapping function $f(x)$from input $x$to output y. It is to determine the specific features or common features to identify an object. In other words, it is to find the hypothesis to determine the target concept. For example, the specific set of features to identify an elephant from all animals.
3. Test – New instances to test the learned model.
4. Formally, Concept learning is defined as:
5. “Given a set of hypotheses, the learner searches through the hypothesis space to identify the best hypothesis that matches the target concept”.
6. Consider the following set of training instances.

Table 3.1: Sample Training Instances

S.No.	Horns	Tail	Tusks	Paws	Fur	Color	Hooves	Size	Elephant
1	No	Short	Yes	No	No	Black	No	Big	Yes
2	Yes	Short	No	No	No	Brown	Yes	Medium	No
3	No	Short	Yes	No	No	Black	No	Medium	Yes
4	No	Long	No	Yes	Yes	White	No	Medium	No
5	No	Short	Yes	Yes	Yes	Black	No	Big	Yes

Here, in this set of training instances, the independent attributes considered are 'Horns', 'Tail', 'Tusks', 'Paws', 'Fur', 'Color', 'Hooves' and 'Size'. The dependent attribute is 'Elephant'. The target concept is to identify the animal to be an Elephant.

Let us now take this example and understand further the concept of hypothesis.

3.4.2 Hypothesis Space

Hypothesis space is the set of all possible hypotheses that approximates the target function f. In other words, the set of all possible approximations of the target function can be defined as hypothesis space. From this set of hypotheses in the hypothesis space, a machine learning algorithm would determine the best possible hypothesis that would best describe the target function or best fit the outputs. Generally, a hypothesis representation language represents a larger hypothesis space.

The set of hypotheses that can be generated by a learning algorithm can be further reduced by specifying a language bias.

The subset of hypothesis space that is consistent with all-observed training instances is called as Version Space. Version space represents the only hypotheses that are used for the classification.

For example, each of the attributes given in Table 3.1 has the following possible set of values.

- Horns – Yes, No
- Tail – Long, Short
- Tusks – Yes, No
- Paws – Yes, No
- Fur – Yes, No
- Color – Brown, Black, White
- Hooves – Yes, No
- Size – Medium, Big

Considering these values for each of the attribute, there are:

$$(2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 2 \times 2) = 384$$

distinct instances covering all the 5 instances in the training dataset.

So, we can generate:

$$(4 \times 4 \times 4 \times 4 \times 4 \times 5 \times 4 \times 4) = 81{,}920$$

distinct hypotheses when including two more values $[?, phi]$for each of the attribute.

However, any hypothesis containing one or more of ϕsymbols represents the empty set of instances; that is, it classifies every instance as a negative instance. Therefore, there will be:

$$(3 \times 3 \times 3 \times 3 \times 3 \times 4 \times 3 \times 3) + 1 = 8{,}749$$

distinct hypotheses by including only ?for each of the attribute and one hypothesis representing the empty set of instances. Thus, the hypothesis space is much larger and hence we need efficient learning algorithms to search for the best hypothesis from the set of hypotheses.

Hypothesis ordering is also important wherein the hypotheses are ordered from the most specific one to the most general one in order to restrict searching the hypothesis space exhaustively.

3.4.3 Heuristic Space Search

Heuristic search is a search strategy that finds an optimized hypothesis/solution to a problem by iteratively improving the hypothesis/solution based on a given heuristic function or a cost measure.

3.4.4 Generalization and Specialization

In order to understand about how we construct this concept hierarchy, let us apply this general principle of generalization/specialization relation. By generalization of the most specific hypothesis and by specialization of the most general hypothesis, the hypothesis space can be searched for an approximate hypothesis that matches all positive instances but does not match any negative instance.

Searching the Hypothesis Space

There are two ways of learning the hypothesis, consistent with all training instances from the large hypothesis space.

1. Specialization – General to Specific learning
2. Generalization – Specific to General learning

Generalization – Specific to General Learning

This learning methodology will search through the hypothesis space for an approximate hypothesis by generalizing the most specific hypothesis

Generalization – Specific to General Learning This learning methodology will search through the hypothesis space for an approximate hypothesis by generalizing the most specific hypothesis.

Example 3.2: Consider the training instances shown in Table 3.1 and illustrate Specific to General Learning.

Solution: We will start from all false or the most specific hypothesis to determine the most restrictive specialization. Consider only the positive instances and generalize the most specific hypothesis. Ignore the negative instances.

This learning is illustrated as follows:

The most specific hypothesis is taken now, which will not classify any instance to true.

h = <φ φ φ φ φ φ φ φ>

Read the first instance $I1$, to generalize the hypothesis h so that this positive instance can be classified by the hypothesis $h1$.

$I1$:	No	Short	Yes	No	No	Black	No	Big	**Yes (Positive instance)**
$h1$ =	<No	Short	Yes	No	No	Black	No	Big>	

When reading the second instance $I2$, it is a negative instance, so ignore it.

$I2$:	Yes	Short	No	No	No	Brown	Yes	Medium	No (Negative instance)
$h2$ =	<No	Short	Yes	No	No	Black	No	Big>	

Similarly, when reading the third instance $I3$, it is a positive instance so generalize $h2$ to $h3$ to accommodate it. The resulting $h3$ is generalized.

$I3$:	No	Short	Yes	No	No	Black	No	Medium	Yes (Positive instance)
$h3$ =	<No	Short	Yes	No	No	Black	No	?>	

Ignore $I4$ since it is a negative instance.

$I4$:	No	Long	No	Yes	Yes	White	No	Medium	**No (Negative instance)**
$h4$ =	<No	Short	Yes	No	No	Black	No	?>	

When reading the fifth instance $I5$, $h4$ is further generalized to $h5$.

$I5$:	No	Short	Yes	Yes	Yes	Black	No	Big	**Yes (Positive instance)**
$h5$ =	<No	Short	Yes	?	?	Black	No	?>	

Now, after observing all the positive instances, an approximate hypothesis $h5$ is generated which can now classify any subsequent positive instance to true.

3.4.5 Hypothesis Space Search by Find-S Algorithm

Specialization – General to Specific Learning This learning methodology will search through the hypothesis space for an approximate hypothesis by specializing the most general hypothesis.

Example 3.3: Illustrate learning by Specialization – General to Specific Learning for the data instances shown in Table 3.1.

Solution: Start from the most general hypothesis which will make true all positive and negative instances.

Initially,

h = <? ? ? ? ? ? ? ?>

h is more general to classify all instances to true.

I1: No Short Yes No No Black No Big Yes (Positive instance)

h1 = <? ? ? ? ? ? ? ?>

I2 : Yes Short No No No Brown Yes Medium No (Negative instance)

h2 = <No ? ? ? ? ? ? ?>
<? ? Yes ? ? ? ? ?>
<? ? ? ? ? Black ? ?>
<? ? ? ? ? ? No ?>
<? ? ? ? ? ? ? Big>

h2 imposes constraints so that it will not classify a negative instance to true.

I3: No Short Yes No No Black No Medium Yes (Positive instance)

h3 = <No ? ? ? ? ? ? ?>
<? ? Yes ? ? ? ? ?>

<? ? ? ? ? Black ? ?>
<? ? ? ? ? ? No ?>
<? ? ? ? ? ? ? Big>

I4: No Long No Yes Yes White No Medium **No (Negative instance)**

h4 = <? ? Yes ? ? ? ? ?>
<? ? ? ? ? Black ? ?>
<? ? ? ? ? ? ? Big>

Remove any hypothesis inconsistent with this negative instance.

I5: No Short Yes Yes Yes Black No Big **Yes (Positive instance)**

h5 = <? ? Yes ? ? ? ? ?>
<? ? ? ? ? Black ? ?>
<? ? ? ? ? ? ? Big>

Thus, $h5$ is the hypothesis space generated which will classify the positive instances to true and negative instances to false.

Find-S algorithm is guaranteed to converge to the most specific hypothesis in Hthat is consistent with the positive instances in the training dataset. Obviously, it will also be consistent with the negative instances. Thus, this algorithm considers only the positive instances and eliminates negative instances while generating the hypothesis. It initially starts with the most specific hypothesis.

Algorithm 3.1: Find-S

Input: Positive instances in the Training dataset

Output: Hypothesis h

1. Initialize $h$to the most specific hypothesis.

$$h = < \phi\phi\phi\phi\phi.... >$$

2. Generalize the initial hypothesis for the first positive instance [Since $h$is more specific].
3. For each subsequent instance:

- If it is a positive instance,
    - Check for each attribute value in the instance with the hypothesis h.
    - If the attribute value is the same as the hypothesis value, then do nothing.
    - Else if the attribute value is different than the hypothesis value, change it to '?' in h.
- Else if it is a negative instance,
    - Ignore it.

Table 3.2: Training Dataset

CGPA	Interactiveness	Practical Knowledge	Communication Skills	Logical Thinking	Interest	Job Offer
≥9	Yes	Excellent	Good	Fast	Yes	Yes
≥9	Yes	Good	Good	Fast	Yes	Yes
≥8	No	Good	Good	Fast	No	No
≥9	Yes	Good	Good	Slow	No	Yes

Solution:

Step 1: Initialize '*h*' to the most specific hypothesis. There are 6 attributes, so for each attribute, we initially fill 'φ' in the initial hypothesis '*h*'.

h = <φ φ φ φ φ φ>

Step 2: Generalize the initial hypothesis for the first positive instance. *I*1 is a positive instance, so generalize the most specific hypothesis '*h*' to include this positive instance. Hence,

*I*1: ≥9 Yes Excellent Good Fast Yes **Positive instance**

h = <≥9 Yes Excellent Good Fast Yes>

Step 3: Scan the next instance *I*2, since *I*2 is a positive instance. Generalize '*h*' to include positive instance *I*2. For each of the non-matching attribute value in '*h*' put a '?' to include this positive instance. The third attribute value is mismatching in '*h*' with *I*2, so put a '?'.

*I*2: ≥9 Yes Good Good Fast Yes **Positive instance**

h = <≥9 Yes ? Good Fast Yes>

Now,scan *I*3. Since it is a negative instance, ignore it. Hence, the hypothesis remains the same without any change after scanning *I*3.

I3: ≥8 No Good Good Fast No **Negative instance**

h = <≥9 Yes ? Good Fast Yes>

Now scan *I*4. Since it is a positive instance, check for mismatch in the hypothesis '*h*' with *I*4. The 5th and 6th attribute value are mismatching, so add '?' to those attributes in '*h*'.

I4: ≥9 Yes Good Good Slow No **Positive instance**

h = <≥9 Yes ? Good ? ?>

Now, the final hypothesis generated with Find-S algorithm is:

h = <≥9 Yes ? Good ? ?>

It includes all positive instances and obviously ignores any negative instance.

Limitations of Find-S Algorithm

1. Find-S algorithm tries to find a hypothesis that is consistent with positive instances, ignoring all negative instances. As long as the training dataset is consistent, the hypothesis found by this algorithm may be consistent.
2. The algorithm finds only one unique hypothesis, wherein there may be many other hypotheses that are consistent with the training dataset.

4.1 INTRODUCTION TO SIMILARITY OR INSTANCE-BASED LEARNING

Similarity-based classifiers use similarity measures to locate the nearest neighbors and classify a test instance which works in contrast with other learning mechanisms such as decision trees or neural networks. Similarity-based learning is also called as Instance-based learning

4.1.1 Differences Between Instance- and Model-based Learning

Instance-based Learning	**Model-based Learning**
Lazy Learners	Eager Learners
Processing of training instances is done only during testing phase	Processing of training instances is done during training phase
No model is built with the training instances before it receives a test instance	Generalizes a model with the training instances before it receives a test instance
Predicts the class of the test instance directly from the training data	Predicts the class of the test instance from the model built
Slow in testing phase	Fast in testing phase
Learns by making many local approximations	Learns by creating global approximation

K-NN Algorithm

Algorithm 4.1: *k*-NN

Inputs: Training dataset T, distance metric d, Test instance t, the number of nearest neighbors k

Output: Predicted class or category

Prediction: For test instance t,

1. For each instance i in T, compute the distance between the test instance t and every other instance i in the training dataset using a distance metric (Euclidean distance).
   [Continuous attributes - Euclidean distance between two points in the plane with coordinates (x_1, y_1) and (x_2, y_2) is given as dist $((x_1, y_1), (x_2, y_2)) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$]
   [Categorical attributes (Binary) - Hamming Distance: If the value of the two instances is same, the distance d will be equal to 0 otherwise $d = 1$.]
2. Sort the distances in an ascending order and select the first k nearest training data instances to the test instance.
3. Predict the class of the test instance by majority voting (if target attribute is discrete valued) or mean (if target attribute is continuous valued) of the k selected nearest instances.

Example 4.1: Consider the student performance training dataset of 8 data instances shown in Table 4.2 which describes the performance of individual students in a course and their CGPA obtained in the previous semesters. The independent attributes are CGPA, Assessment and Project. The target variable is 'Result' which is a discrete valued variable that takes two values 'Pass' or 'Fail'. Based on the performance of a student, classify whether a student will pass or fail in that course.

Table 4.2: Training Dataset *T*

S.No.	CGPA	Assessment	Project Submitted	Result
1.	9.2	85	8	Pass
2.	8	80	7	Pass
3.	8.5	81	8	Pass

(Continued)

S.No.	CGPA	Assessment	Project Submitted	Result
4.	6	45	5	Fail
5.	6.5	50	4	Fail
6.	8.2	72	7	Pass
7.	5.8	38	5	Fail
8.	8.9	91	9	Pass

Solution: Given a test instance (6.1, 40, 5) and a set of categories {Pass, Fail} also called as classes, we need to use the training set to classify the test instance using Euclidean distance.

The task of classification is to assign a category or class to an arbitrary instance.

Assign $k = 3$.

Step 1: Calculate the Euclidean distance between the test instance (6.1, 40, and 5) and each of the training instances as shown in Table 4.3.

Step 1: Calculate the Euclidean distance between the test instance (6.1, 40, and 5) and each of the training instances as shown in Table 4.3.

Table 4.3: Euclidean Distance

S.No.	CGPA	Assessment	Project Submitted	Result	Euclidean Distance
1.	9.2	85	8	Pass	$\sqrt{(9.2-6.1)^2+(85-40)^2+(8-5)^2}$ $= 45.2063$
2.	8	80	7	Pass	$\sqrt{(8-6.1)^2+(80-40)^2+(7-5)^2}$ $= 40.09501$
3.	8.5	81	8	Pass	$\sqrt{(8.5-6.1)^2+(81-40)^2+(8-5)^2}$ $= 41.17961$
4.	6	45	5	Fail	$\sqrt{(6-6.1)^2+(45-40)^2+(5-5)^2}$ $= 5.001$
5.	6.5	50	4	Fail	$\sqrt{(6.5-6.1)^2+(50-40)^2+(4-5)^2}$ $= 10.05783$
6.	8.2	72	7	Pass	$\sqrt{(8.2-6.1)^2+(72-40)^2+(7-5)^2}$ $= 32.13114$
7.	5.8	38	5	Fail	$\sqrt{(5.8-6.1)^2+(38-40)^2+(5-5)^2}$ $= 2.022375$
8.	8.9	91	9	Pass	$\sqrt{(8.9-6.1)^2+(91-40)^2+(9-5)^2}$ $= 51.23319$

Step 2: Sort the distances in the ascending order and select the first 3 nearest training data instances to the test instance. The selected nearest neighbors are shown in Table 4.4.

Table 4.4: Nearest Neighbors

Instance	Euclidean Distance	Class
4	5.001	Fail
5	10.05783	Fail
7	2.022375	Fail

Here, we take the 3 nearest neighbors as instances 4, 5 and 7 with smallest distances.

Step 3: Predict the class of the test instance by majority voting.

The class for the test instance is predicted as 'Fail'.

Weighted KNN Algorithm

4.3 WEIGHTED *K*-NEAREST-NEIGHBOR ALGORITHM

The Weighted *k*-NN is an extension of *k*-NN. It chooses the neighbors by using the weighted distance. The *k*-Nearest Neighbor (*k*-NN) algorithm has some serious limitations as its performance is solely dependent on choosing the *k* nearest neighbors, the distance metric used and the decision rule. However, the principle idea of Weighted *k*-NN is that *k* closest neighbors to the test instance are assigned a higher weight in the decision as compared to neighbors that are farther away from the test instance. The idea is that weights are inversely proportional to distances.

The selected *k* nearest neighbors can be assigned uniform weights, which means all the instances in each neighborhood are weighted equally or weights can be assigned by the inverse of their distance. In the second case, closer neighbors of a query point will have a greater influence than neighbors which are further away.

Algorithm 4.2: Weighted *k*-NN

Inputs: Training dataset '*T*', Distance metric '*d*', Weighting function *w*(*i*), Test instance '*t*', the number of nearest neighbors '*k*'

(Continued)

Output: Predicted class or category

Prediction: For test instance *t*,

1. For each instance '*i*' in Training dataset *T*, compute the distance between the test instance *t* and every other instance '*i*' using a distance metric (Euclidean distance).
   [Continuous attributes - Euclidean distance between two points in the plane with coordinates (x_1, y_1) and (x_2, y_2) is given as dist $((x_1, y_1), (x_2, y_2)) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$]
   [Categorical attributes (Binary) - Hamming Distance: If the values of two instances are the same, the distance *d* will be equal to 0. Otherwise $d = 1$.]
2. Sort the distances in the ascending order and select the first '*k*' nearest training data instances to the test instance.
3. Predict the class of the test instance by weighted voting technique (Weighting function *w*(*i*)) for the *k* selected nearest instances:
   - Compute the inverse of each distance of the '*k*' selected nearest instances.
   - Find the sum of the inverses.
   - Compute the weight by dividing each inverse distance by the sum. (Each weight is a vote for its associated class).
   - Add the weights of the same class.
   - Predict the class by choosing the class with the maximum vote.

Example 4.2: Consider the same training dataset given in Table 4.1. Use Weighted k-NN and determine the class.

Solution:

Step 1: Given a test instance (7.6, 60, 8) and a set of classes {Pass, Fail}, use the training dataset to classify the test instance using Euclidean distance and weighting function.

Assign $k = 3$. The distance calculation is shown in Table 4.5.

Table 4.5: Euclidean Distance

S.No.	CGPA	Assessment	Project Submitted	Result	Euclidean Distance
1.	9.2	85	8	Pass	$\sqrt{(9.2-7.6)^2+(85-60)^2+(8-8)^2}$ $= 25.05115$
2.	8	80	7	Pass	$\sqrt{(8-7.6)^2+(80-60)^2+(7-8)^2}$ $= 20.02898$
3.	8.5	81	8	Pass	$\sqrt{(8.5-7.6)^2+(81-60)^2+(8-8)^2}$ $= 21.01928$

(Continued)

S.No.	CGPA	Assessment	Project Submitted	Result	Euclidean Distance
4.	6	45	5	Fail	$\sqrt{(6-7.6)^2+(45-60)^2+(5-8)^2}$ $= 15.38051$
5.	6.5	50	4	Fail	$\sqrt{(6.5-7.6)^2+(50-60)^2+(4-8)^2}$ $= 10.82636$
6.	8.2	72	7	Pass	$\sqrt{(8.2-7.6)^2+(72-60)^2+(7-8)^2}$ $= 12.05653$
7.	5.8	38	5	Fail	$\sqrt{(5.8-7.6)^2+(38-60)^2+(5-8)^2}$ $= 22.27644$
8.	8.9	91	9	Pass	$\sqrt{(8.9-7.6)^2+(91-60)^2+(9-8)^2}$ $= 31.04336$

Step 2: Sort the distances in the ascending order and select the first 3 nearest training data instances to the test instance. The selected nearest neighbors are shown in Table 4.6.

Table 4.6: Nearest Neighbors

Instance	Euclidean Distance	Class
4	15.38051	Fail
5	10.82636	Fail
6	12.05653	Pass

Step 3: Predict the class of the test instance by weighted voting technique from the 3 selected nearest instances.

- Compute the inverse of each distance of the 3 selected nearest instances as shown in Table 4.7.

Table 4.7: Inverse Distance

Instance	Euclidean Distance	Inverse Distance	Class
4	15.38051	0.06502	Fail
5	10.82636	0.092370	Fail
6	12.05653	0.08294	Pass

- Find the sum of the inverses.

  Sum = 0.06502 + 0.092370 + 0.08294 = 0.24033

- Compute the weight by dividing each inverse distance by the sum as shown in Table 4.8.

Table 4.8: Weight Calculation

Instance	Euclidean Distance	Inverse Distance	Weight = Inverse distance/Sum	Class
4	15.38051	0.06502	0.270545	Fail
5	10.82636	0.092370	0.384347	Fail
6	12.05653	0.08294	0.345109	Pass

- Add the weights of the same class.

  Fail = 0.270545 + 0.384347 = 0.654892

  Pass = 0.345109

- Predict the class by choosing the class with the maximum vote.

The class is predicted as **'Fail'**.

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School of Computer Science and Technology

Regression Analysis(5.1,5.2,5.3)

What is Regression Analysis?

- Regression analysis is a set of statistical methods used for the estimation of relationships between a dependent variable and one or more independent variables. It can be utilized to assess the strength of the relationship between variables and for modeling the future relationship between them.

- For example: Y=F(X)
- HERE, Y IS CALLED DEPENDENT VARIABLE AND X IS CALLED INDEPENDENT VARIABLE

Focus of Regression Analysis

- What is the relationship between the variables?
- What is the strength of the relationships?
- What is the nature of the relationship such as linear or non-linear?
- What is the relevance of the attributes?
- What is the contribution of each attribute?

Applications of Regression Analysis

There are many applications of regression analysis. Some of the applications of regressions include predicting:

- Sales of a goods or services
- Value of bonds in portfolio management
- Premium on insurance companies
- Yield of crops in agriculture
- Prices of real estate

Regression and Correlation

- Correlation quantifies how strong and in which direction the two variables are related. It is being measured using the correlation coefficient, which goes to the extent of negative one and positive one.

- Positive Correlation: In this case, if the value of one of them goes up, then the value of the other one also goes up.
- Negative Correlation: That is, when one variable is high, the other is low, and vice versa.
- No Correlation: Not statistically significantly correlated: partners, financial wealth, and life satisfaction.

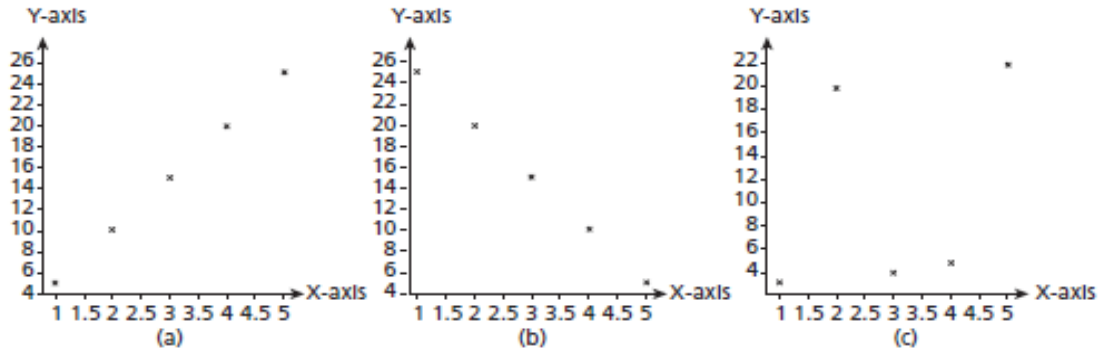


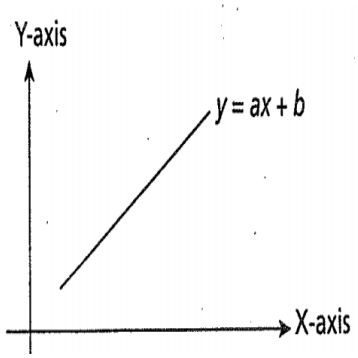
Figure 5.1: Examples of (a) Positive Correlation (b) Negative Correlation (c) Random Points with No Correlation

Regression and Causation

- Causation means one variable directly affects another (cause → effect).
- Correlation does NOT imply causation.
- Just because two variables are related does not mean one causes the other.
- Examples:
- If x causes y, then a change in x leads to a change in y.
- But: Higher temperature and cold drink sales may be related,
- However, other factors (marketing, season, location) may also influence sales.

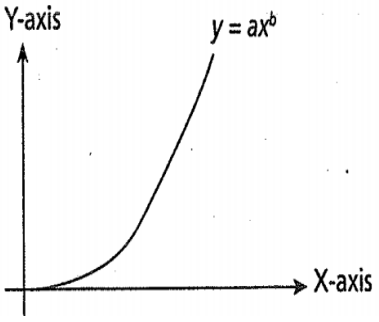
Linear and Non-Linear Relationships

- A. Linear Relationship
- A relationship is linear if it forms a straight line.
- General form:
- y=ax+b
- When x increases, y increases (or decreases) at a constant rate.
- Example:
- Salary increases steadily with years of experience.



Non-Linear Relationship

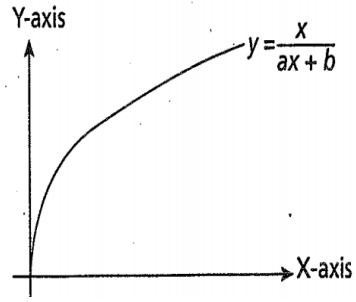
A relationship is non-linear if it is not a straight line.
The rate of change is not constant.
Common type:
1. Exponential Function: y=a^b
Growth becomes faster over time.
Example:
Population growth, compound interest



2. Power Function

$$y = \frac{x}{ax+b}$$

Curve increases but slows down.
Example:
Efficiency or saturation effects.



Types of Regression

- **Linear Regression**

It is a type of regression where a line is fitted upon given data for finding the linear relationship between one independent variable and one dependent variable to describe relationships.

- **Multiple Regression**

It is a type of regression where a line is fitted for finding the linear relationship between two or more independent variables and one dependent variable to describe relationships among variables.

- **Polynomial Regression**

It is a type of non-linear regression method of describing relationships among variables where Nth degree polynomial is used to model the relationship between one independent variable and one dependent variable. Polynomial multiple regression is used to model two or more independent variables and one dependent variable.

- **Logistic Regression**

It is used for predicting categorical variables that involve one or more independent variables and one dependent variable. This is also known as a binary classifier.

- **Lasso and Ridge Regression Methods**

These are special variants of regression method where regularization methods are used to limit the number and size of coefficients of the independent variables.

Types of Regression Methods

The classification of regression methods is shown in Figure 5.3.

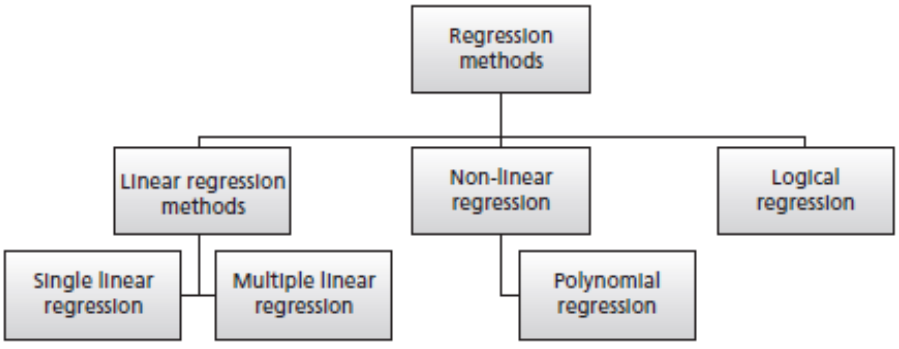


Figure 5.3: Types of Regression Methods

Limitations of Regression Method

- **Outliers –** Outliers are abnormal data. It can bias the outcome of the regression model, as outliers push the regression line towards it.
- **Number of cases –** The ratio of independent and dependent variables should be at least 20 : 1. For every explanatory variable, there should be at least 20 samples. Atleast five samples are required in extreme cases.
- **Missing data –** Missing training data can make the model unfit for the sampled data.
- **Multicollinearity –** If exploratory variables are highly correlated (0.9 and above), the regression is vulnerable to bias. Singularity leads to perfect correlation of 1. The remedy is to remove exploratory variables that exhibit correlation more than 1. If there is a tie, then the tolerance (1 − R squared) is used to eliminate variables that have the greatest value.

Linear Regression

- Linear Regression is a fundamental supervised learning algorithm used to model the relationship between a dependent variable and one or more independent variables. It predicts continuous values by fitting a straight line that best represents the data.
- It assumes that there is a linear relationship between the input and output
- Uses a best-fit line to make predictions
- Commonly used in forecasting, trend analysis, and predictive modelling

Linear Regression (Continued)

- The assumptions of linear regression are listed as follows:
- The observations (y) are random and are mutually independent.
- The difference between the predicted and true values is called an error. The error is also mutually independent with the same distributions such as normal distribution with zero mean and constant variables.
- The distribution of the error term is independent of the joint distribution of explanatory variables.
- The unknown parameters of the regression models are constants.

$$y = a_0 + a_1 \times x + e$$

Linear Regression Model

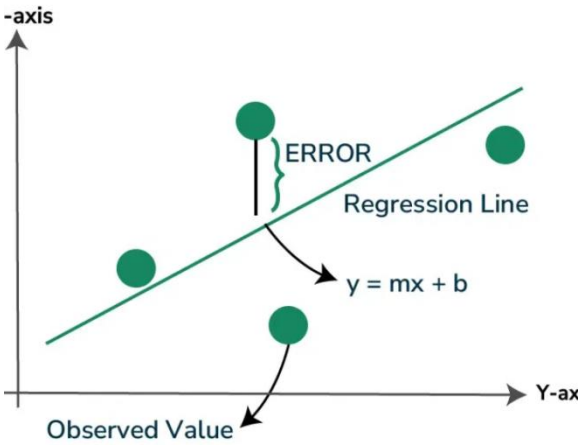
$$J(a_1, a_0) = \sum_{i=1}^{n}[y_i - f(x_i)]^2$$

$$= \sum_{i=1}^{n}[y_i - (a_0 + a_1 x_i)]^2$$

$$a_1 = \frac{(\overline{xy}) - (\overline{x})(\overline{y})}{(\overline{x^2}) - (\overline{x})^2} \quad (5.9)$$

And the value of a_0 is given as follows:

$$a_0 = (\overline{y}) - a_1 \times \overline{x} \quad (5.10)$$



Linear Regression Example

Example 5.1: Let us consider an example where the five weeks' sales data (in Thousands) is given as shown below in Table 5.1. Apply linear regression technique to predict the 7th and 9th month sales.

Table 5.1: Sample Data

x_i (Week)	y_i (Sales in Thousands)
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8

Table 5.2: Computation Table

x_i	y_i	$(x_i)^2$	$x_i \times y_i$
1	1.2	1	1.2
2	1.8	4	3.6
3	2.6	9	7.8
4	3.2	16	12.8
5	3.8	25	19
Sum = 15 Average of (x_i) $= \overline{x} = \frac{15}{5}$ $= 3$	Sum = 12.6 Average of (y_i) $= \overline{y} = \frac{12.6}{5}$ $= 2.52$	Sum = 55 Average of (x_i^2) $= \overline{x_i^2} = \frac{55}{5}$ $= 11$	Sum = 44.4 Average of $(x_i \times y_i)$ $= \overline{xy} = \frac{44.4}{5}$ $= 8.88$

Let us compute the slope and intercept now using Eq. (5.9) as:

$$a_1 = \frac{8.88 - 3(2.52)}{11 - 3^2} = 0.66$$

$$a_0 = 2.52 - 0.66 \times 3 = 0.54$$

The fitted line is shown in Figure 5.5.

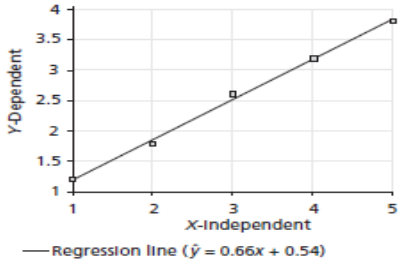


Figure 5.5: Linear Regression Model Constructed

Let us model the relationship as $y = a_0 + a_1 \times x$. Therefore, the fitted line for the above data is: $y = 0.54 + 0.66 \times x$.

The predicted 7th week sale would be (when $x = 7$), $y = 0.54 + 0.66 \times 7 = 5.16$ and the 12th month, $y = 0.54 + 0.66 \times 12 = 8.46$. All sales are in thousands.

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Decision Tree (6.1,6.2)

Decision Tree Learning Model

- Decision tree learning model, one of the most popular supervised predictive learning models, classifies data instances with high accuracy and consistency. The model performs an inductive inference that reaches a general conclusion from observed examples. This model is variably used for solving complex classification applications.

- Decision tree is a concept tree which summarizes the information contained in the training dataset in the form of a tree structure. Once the concept model is built, test data can be easily classified.

Structure of a Decision Tree

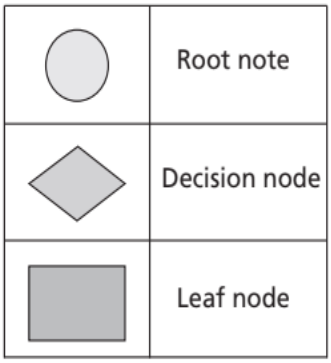


Figure 6.1: Nodes in a Decision Tree

BUILDING THE TREE

Goal: Construct a decision tree with the given training dataset. The tree is constructed in a top-down fashion. It starts from the root node. At every level of tree construction, we need to find the best split attribute or best decision node among all attributes. This process is recursive and continued until we end up in the last level of the tree or finding a leaf node which cannot be split further. The tree construction is complete when all the test conditions lead to a leaf node. The leaf node contains the target class or output of classification.

Output: Decision tree representing the complete hypothesis space.

KNOWLEDGE INFERENCE OR CLASSIFICATION

Goal Given a test instance, infer to the target class it belongs to.

Classification Inferring the target class for the test instance or object is based on inductive inference on the constructed decision tree. In order to classify an object, we need to start traversing the tree from the root. We traverse as we evaluate the test condition on every decision node with the test object attribute value and walk to the branch corresponding to the test's outcome. This process is repeated until we end up in a leaf node which contains the target class of the test object.

Output Target label of the test instance.

Advantages of Decision Trees

- Easy to model and interpret
- Simple to understand
- The input and output attributes can be discrete or continuous predictor variables.
- Can model a high degree of nonlinearity in the relationship between the target variables and the predictor variables
- Quick to train
-

Some of the issues that generally arise with a decision tree learning are that:

Disadvantages of Decision Trees

- It is difficult to determine how deeply a decision tree can be grown or when to stop growing it.
- If training data has errors or missing attribute values, then the decision tree constructed may become unstable or biased.
- If the training data has continuous valued attributes, handling it is computationally complex and has to be discretized.
- A complex decision tree may also be over-fitting with the training data.
- Decision tree learning is not well suited for classifying multiple output classes.
- Learning an optimal decision tree is also known to be NP-complete.

Fundamentals of Entropy

- Entropy is the amount of uncertainty or randomness in the outcome of a random variable or an event. Moreover, entropy describes about the homogeneity of the data instances. The best feature is selected based on the entropy value. For example, when a coin is flipped, head or tail are the two outcomes, hence its entropy is lower when compared to rolling a dice which has got six outcomes.

- Higher the entropy → Higher the uncertainty
- Lower the entropy → Lower the uncertainty

$$\text{Entropy_Info}(P) = \text{Entropy_Info}(P_1 \ldots P_n)$$
$$= -(P_1 \log_2(P_1) + P_2 \log_2(P_2) + \ldots\ldots + P_n \log_2(P_n))$$

Decision Tree Induction Algorithms

- There are many decision tree algorithms, such as ID3, C4.5, CART, CHAID, QUEST, GUIDE, CRUISE, and CTREE, that are used for classification in real-time environment. The most commonly used decision tree algorithms are ID3 (Iterative Dichotomizer 3), developed by J.R Quinlan in 1986.

- The accuracy of the tree constructed depends upon the selection of the best split attribute. Different algorithms are used for building decision trees which use different measures to decide on the splitting criterion. Algorithms such as ID3, C4.5 and CART are popular algorithms used in the construction of decision trees. The algorithm ID3 uses 'Information Gain' as the splitting criterion.

ID3 Tree Construction

Algorithm 6.2: Procedure to Construct a Decision Tree using ID3

1. Compute Entropy_Info Eq. (6.8) for the whole training dataset based on the target attribute.
2. Compute Entropy_Info Eq. (6.9) and Information_Gain Eq. (6.10) for each of the attribute in the training dataset.
3. Choose the attribute for which entropy is minimum and therefore the gain is maximum as the best split attribute.
4. The best split attribute is placed as the root node.
5. The root node is branched into subtrees with each subtree as an outcome of the test condition of the root node attribute. Accordingly, the training dataset is also split into subsets.
6. Recursively apply the same operation for the subset of the training set with the remaining attributes until a leaf node is derived or no more training instances are available in the subset.

$$\text{Information_Gain}(A) = \text{Entropy_Info}(T) - \text{Entropy_Info}(T, A) \quad (6.10)$$

It can be noted that as entropy increases, information gain decreases. They are inversely proportional to each other.

Example of ID3 Construction

Example 6.3: Assess a student's performance during his course of study and predict whether a student will get a job offer or not in his final year of the course. The training dataset T consists of 10 data instances with attributes such as 'CGPA', 'Interactiveness', 'Practical Knowledge' and 'Communication Skills' as shown in Table 6.3. The target class attribute is the 'Job Offer'.

Table 6.3: Training Dataset T

S.No.	CGPA	Interactiveness	Practical Knowledge	Communication Skills	Job Offer
1.	≥9	Yes	Very good	Good	Yes
2.	≥8	No	Good	Moderate	Yes
3.	≥9	No	Average	Poor	No
4.	<8	No	Average	Good	No
5.	≥8	Yes	Good	Moderate	Yes
6.	≥9	Yes	Good	Moderate	Yes
7.	<8	Yes	Good	Poor	No
8.	≥9	No	Very good	Good	Yes
9.	≥8	Yes	Good	Good	Yes
10.	≥8	Yes	Average	Good	Yes

Step 1:
Calculate the Entropy for the target class 'Job Offer'.

$$= -\left[\frac{7}{10}\log_2\frac{7}{10} + \frac{3}{10}\log_2\frac{3}{10}\right] = -(-0.3599 + -0.5208) = 0.8807$$

Iteration 1:

Step 2:
Calculate the Entropy_Info and Gain(Information_Gain) for each of the attribute in the training dataset.

Table 6.4 shows the number of data instances classified with Job Offer as Yes or No for the attribute CGPA.

CGPA	Job Offer = Yes	Job Offer = No	Total	Entropy
≥9	3	1	4	
≥8	4	0	4	0

CGPA	Job Offer = Yes	Job Offer = No	Total	Entropy
<8	0	2	2	0

Entropy_Info(T, CGPA)

$$\text{Entropy_Info}(T, \text{CGPA}) = \frac{4}{10}\left[-\frac{3}{4}\log_2\frac{3}{4} - \frac{1}{4}\log_2\frac{1}{4}\right] + \frac{4}{10}\left[-\frac{4}{4}\log_2\frac{4}{4} - \frac{0}{4}\log_2\frac{0}{4}\right] + \frac{2}{10}\left[-\frac{0}{2}\log_2\frac{0}{2} - \frac{2}{2}\log_2\frac{2}{2}\right]$$

$$= \frac{4}{10}(0.3111 + 0.4997) + 0 + 0$$

$$= 0.3243$$

$$\text{Gain (CGPA)} = 0.8807 - 0.3243$$

$$= 0.5564$$

Table 6.5 shows the number of data instances classified with Job Offer as Yes or No for the attribute Interactiveness.

Interactiveness	Job Offer = Yes	Job Offer = No	Total	Entropy
YES	5	1	6	
NO	2	2	4	

Entropy_Info(T,Interactiveness)

$$\text{Entropy_Info}(T, \text{Interactiveness}) = \frac{6}{10}\left[-\frac{5}{6}\log_2\frac{5}{6} - \frac{1}{6}\log_2\frac{1}{6}\right] + \frac{4}{10}\left[-\frac{2}{4}\log_2\frac{2}{4} - \frac{2}{4}\log_2\frac{2}{4}\right]$$

$$= \frac{6}{10}(0.2191 + 0.4306) + \frac{4}{10}(0.4997 + 0.4997)$$

$$= 0.3898 + 0.3998 = 0.7896$$

$$\text{Gain(Interactiveness)} = 0.8807 - 0.7896$$

$$= 0.0911$$

Entropy Information for Practical Knowledge

Practical Knowledge	Job Offer = Yes	Job Offer = No	Total	Entropy
Very Good	2	0	2	0
Average	1	2	3	

Practical Knowledge	Job Offer = Yes	Job Offer = No	Total	Entropy
Good	4	1	5	

Entropy_Info(T, Practical Knowledge)

$$= \frac{2}{10}\left[-\frac{2}{2}\log_2\frac{2}{2} - \frac{0}{2}\log_2\frac{0}{2}\right] + \frac{3}{10}\left[-\frac{1}{3}\log_2\frac{1}{3} - \frac{2}{3}\log_2\frac{2}{3}\right] + \frac{5}{10}\left[-\frac{4}{5}\log_2\frac{4}{5} - \frac{1}{5}\log_2\frac{1}{5}\right]$$

$$= \frac{2}{10}(0) + \frac{3}{10}(0.5280 + 0.3897) + \frac{5}{10}(0.2574 + 0.4641)$$

$$= 0 + 0.2753 + 0.3608$$

$$= 0.6361$$

$$\text{Gain(Practical Knowledge)} = 0.8807 - 0.6361$$

$$= 0.2446$$

shows the number of data instances classified with Job Offer as Yes or No for the attribute Communication Skills.

Entropy Information for Communication Skills

Communication Skills	Job Offer = Yes	Job Offer = No	Total
Good	4	1	5
Moderate	3	0	3
Poor	0	2	2

Entropy_Info(T, Communication Skills)

$$\text{Entropy_Info}(T, \text{Communication Skills})$$

$$= \frac{5}{10}\left[-\frac{4}{5}\log_2\frac{4}{5} - \frac{1}{5}\log_2\frac{1}{5}\right] + \frac{3}{10}\left[-\frac{3}{3}\log_2\frac{3}{3} - \frac{0}{3}\log_2\frac{0}{3}\right] + \frac{2}{10}\left[-\frac{0}{2}\log_2\frac{0}{2} - \frac{2}{2}\log_2\frac{2}{2}\right]$$

$$= \frac{5}{10}(0.5280 + 0.3897) + \frac{3}{10}(0) + \frac{2}{10}(0)$$

$$= 0.3609$$

$$\text{Gain(Communication Skills)} = 0.8813 - 0.36096$$

$$= 0.5203$$

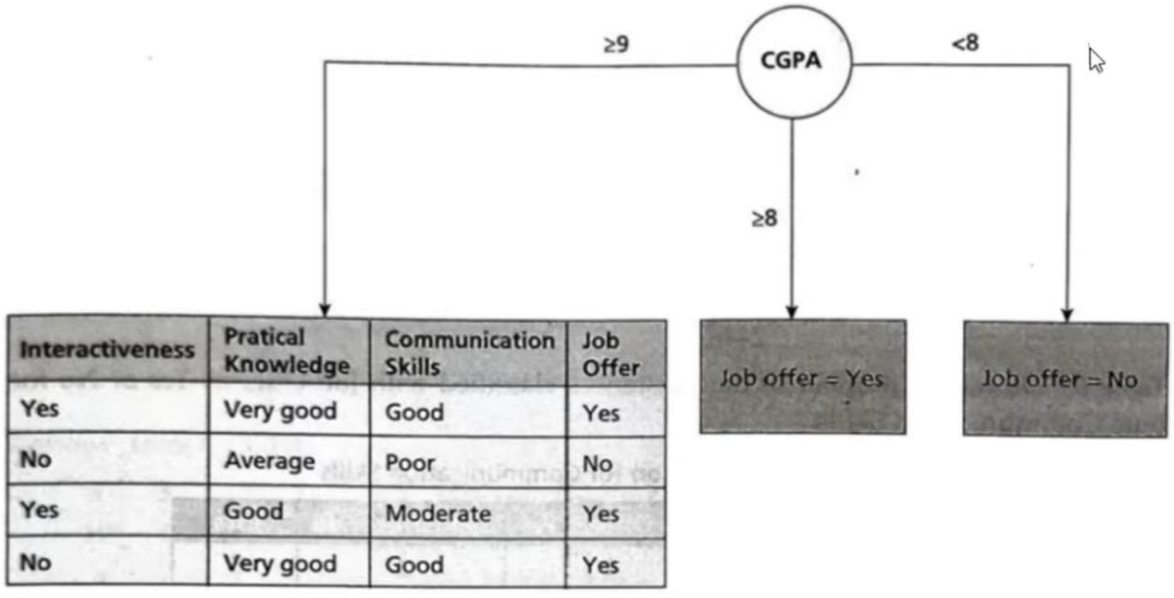
The Gain calculated for all the attributes is shown in Table 6.8:

Table 6.8: Gain

Attributes	Gain
CGPA	0.5564
Interactivityness	0.0911
Practical Knowledge	0.2246
Communication Skills	0.5203

Step 3: From Table 6.8, choose the attribute for which entropy is minimum and therefore the gain is maximum as the best split attribute.

The best split attribute is CGPA since it has the maximum gain. So, we choose CGPA as the root node. There are three distinct values for CGPA with outcomes ≥9, ≥8 and <8. The entropy value is 0 for ≥8 and <8 with all instances classified as Job Offer = Yes for ≥8 and Job Offer = No for <8. Hence, both ≥8 and <8 end up in a leaf node. The tree grows with the subset of instances with CGPA ≥9 as shown in Figure 6.3.



Interactiveness	Pratical Knowledge	Communication Skills	Job Offer
Yes	Very good	Good	Yes
No	Average	Poor	No
Yes	Good	Moderate	Yes
No	Very good	Good	Yes

Figure 6.3: Decision Tree After Iteration 1

Iteration 2:

In this iteration, the same process of computing the Entropy_Info and Gain are repeated with the subset of training set. The subset consists of 4 data instances as shown in the above Figure 6.3.

Entropy_Info(T) = Entropy_Info(3, 1) =

$$= -\left[\frac{3}{4}\log_2\frac{3}{4} + \frac{1}{4}\log_2\frac{1}{4}\right]$$
$$= -(-0.3111 + -0.4997)$$
$$= 0.8108$$

$$\text{Entropy_Info}(T, \text{Interactiveness}) = \frac{2}{4}\left[-\frac{2}{2}\log_2\frac{2}{2} - \frac{0}{2}\log_2\frac{0}{2}\right] + \frac{2}{4}\left[-\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2}\right]$$
$$= 0 + 0.4997$$

$$\text{Gain(Interactiveness)} = 0.8108 - 0.4997$$
$$= 0.3111$$

Entropy_Info(T, Practical Knowledge)

$$\text{Entropy_Info}(T, \text{Practical Knowledge})$$
$$= \frac{2}{4}\left[-\frac{2}{2}\log_2\frac{2}{2} - \frac{0}{2}\log_2\frac{0}{2}\right] + \frac{1}{4}\left[-\frac{0}{1}\log_2\frac{0}{1} - \frac{1}{1}\log_2\frac{1}{1}\right] + \frac{1}{4}\left[-\frac{1}{1}\log_2\frac{1}{1} - \frac{0}{1}\log_2\frac{0}{1}\right]$$
$$= 0$$

$$\text{Gain(Practical Knowledge)} = 0.8108$$

$$\text{Entropy_Info}(T, \text{Communication Skills})$$
$$= \frac{2}{4}\left[-\frac{2}{2}\log_2\frac{2}{2} - \frac{0}{2}\log_2\frac{0}{2}\right] + \frac{1}{4}\left[-\frac{0}{1}\log_2\frac{0}{1} - \frac{1}{1}\log_2\frac{1}{1}\right] + \frac{1}{4}\left[-\frac{1}{1}\log_2\frac{1}{1} - \frac{0}{1}\log_2\frac{0}{1}\right]$$
$$= 0$$

$$\text{Gain(Communication Skills)} = 0.8108$$

Total Gain

Attributes	Gain
Interactivities	0.3111
Practical Knowledge	0.8108
Communication Skills	0.8108

Here, both the attributes 'Practical Knowledge' and 'Communication Skills' have the same Gain. So, we can either construct the decision tree using 'Practical Knowledge' or 'Communication Skills'.

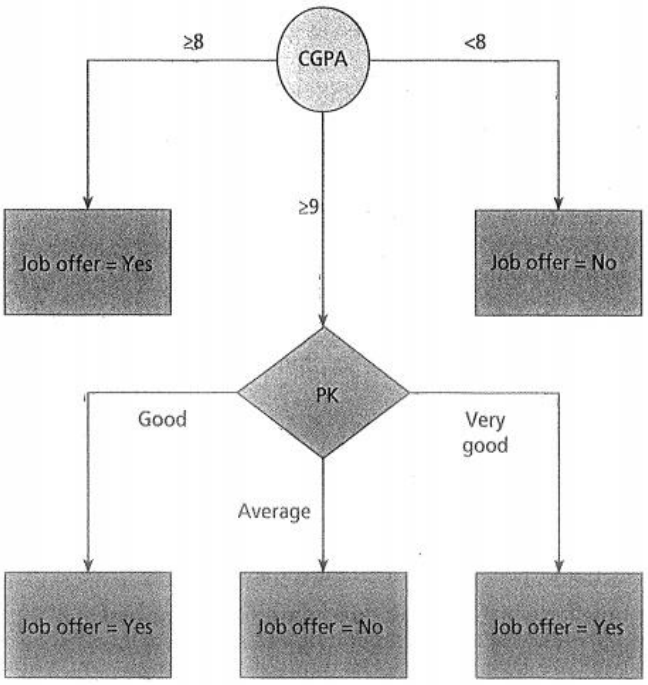


Figure 6.4: Final Decision Tree

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Introduction to Bayesian Learning

Bayesian Learning is a probability-based machine learning approach that uses Bayes' Theorem to make predictions and decisions. It helps in updating the probability of a hypothesis whenever new evidence or information is available.

In simple words, Bayesian learning learns from past data and improves predictions when new data is received.

It is based on the idea:

"Initial belief + New evidence = Updated belief"

Bayesian learning calculates probabilities for different outcomes and selects the outcome with the highest probability.

Bayes' Theorem

Bayes' Theorem is a probability theorem used to find the probability of an event after getting new information or evidence. It helps in updating the existing probability based on additional knowledge.

In simple words, Bayes' theorem tells us how likely something is when some related information is already known.

It is one of the most important concepts in probability-based learning and machine learning.

The formula for Bayes' theorem is:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

Where:

- $P(A|B)$ = Probability of event A given that event B has occurred (Posterior Probability)
- $P(B|A)$ = Probability of event B given that event A has occurred (Likelihood)
- $P(A)$ = Initial probability of event A (Prior Probability)
- $P(B)$ = Total probability of event B (Evidence)

Bayes' theorem works by combining prior knowledge with new evidence to produce a better and updated probability.

Example

Suppose a boy usually participates in volleyball tournaments. If we know that the boy is sick, Bayes' theorem helps us calculate the probability that he will still participate in the tournament.

Thus, Bayes' theorem is very useful in decision-making and prediction problems.

Applications of Bayes' Theorem

- Spam email filtering
- Medical diagnosis
- Weather forecasting
- Machine learning algorithms
- Recommendation systems

Conclusion

Bayes' theorem is an important statistical method used to calculate conditional probabilities. It helps in improving predictions by using prior information and new evidence together.

Term	**Formula**	**Simple Meaning**
Posterior Probability	(P(A|B))	Final updated probability
Likelihood	(P(B|A))	Probability of evidence if event is true
Prior Probability	(P(A))	Initial probability before evidence
Evidence	(P(B))	Total probability of evidence

Very Easy Memory Trick

Prior

☞ What we believed BEFORE.

Evidence

☞ New information we got.

Likelihood

☞ How strongly the evidence supports the event.

Posterior

☞ Final updated belief AFTER using evidence.

Consider a boy who has a volleyball tournament on the next day, but today he feels sick. It is unusual that there is only a 40% chance he would fall sick since he is a healthy boy. Now, find the probability of the boy participating in the tournament. The boy is very much interested in volleyball, so there is a 90% probability that he would participate in tournaments and 20% that he will fall sick given that he participates in the tournament.

Bayes' theorem formula:

$$P(T|S) = \frac{P(S|T) \times P(T)}{P(S)}$$

Where:

- T = Boy participates in tournament
- S = Boy is sick

1. Prior Probability

$$P(T) = 0.9$$

Meaning

☞ Probability of the boy participating **before knowing he is sick**.

In Our Example

The boy loves volleyball.

So initially:

☞ There is a 90% chance he participates.

This is called:

✅ Prior Probability

Because it is our belief BEFORE new information.

2. Evidence

$$P(S) = 0.4$$

Meaning

☞ Probability that the boy is sick.

In Our Example

We know:

☞ There is a 40% chance the boy becomes sick.

This sickness information is the evidence.

So:

✅ Evidence = Probability of sickness

3. Likelihood

$$P(S|T) = 0.2$$

Meaning

☞ Probability the boy is sick GIVEN that he participates.

In Our Example

Suppose the boy still goes to tournaments even sometimes when sick.

So:

☞ Among participation situations, sickness happens 20% of the time.

This tells how strongly sickness and participation are related.

So this is called:

✅ Likelihood

4. Posterior Probability

$$P(T|S)$$

Meaning

☞ Probability the boy participates AFTER knowing he is sick.

This is the final answer we calculate using Bayes theorem.

After calculation:

$$P(T|S) = \frac{0.2 \times 0.9}{0.4}$$

$$P(T|S) = 45\%$$

Meaning:

☞ Even if the boy is sick, there is still a 45% chance he participates.

This updated probability is called:

✅ Posterior Probability

Probability the boy participates **given that he is sick**

$$P(T|S)$$

Meaning:

☞ “If we already know he is sick, what is the chance he still plays?”

Use Bayes’ Theorem

$$P(T|S) = \frac{P(S|T) \times P(T)}{P(S)}$$

Now put the values:

$$P(T|S) = \frac{0.2 \times 0.9}{0.4}$$

Final Answer

$$P(T|S) = 0.45 = 45\%$$

So there is a **45% chance** the boy will still participate in the tournament even though he is sick.

Naïve Bayes Theorem

Naïve Bayes is a supervised machine learning algorithm used for classification problems. It is based on Bayes’ Theorem and is mainly used to predict the class or category of data.

It is called “Naïve” because it assumes that all attributes are independent of each other.

For example, while predicting whether a student gets a job offer, Naïve Bayes assumes that:

- CGPA,
- Communication Skills,
- Practical Knowledge, and

- Interactiveness

Definition

Naïve Bayes is a probabilistic classification algorithm that uses Bayes' theorem with the assumption that features are conditionally independent.

Naïve Bayes Formula

In Naïve Bayes, multiple attributes are considered:

$$P(C|X) = P(x_1|C) \times P(x_2|C) \times P(x_3|C) \times \cdots \times P(C)$$

Where:

- C = Class label (Yes/No)
- X = Input attributes
- x_1, x_2, x_3 = Features

Q. Assess a student's performance using Naïve Bayes algorithm with the dataset provided in Table 8.1. Predict whether a student gets a job offer or not in his final year of the course.

CGPA	Interactiveness	Practical Knowledge	Communication Skills	Job Offer
≥9	Yes	Very Good	Good	Yes
≥8	No	Good	Moderate	Yes
≥9	No	Average	Poor	No
<8	No	Average	Good	No
≥8	Yes	Good	Moderate	Yes
≥9	Yes	Good	Moderate	Yes
<8	Yes	Good	Poor	No
≥9	No	Very Good	Good	Yes

CGPA	Interactiveness	Practical Knowledge	Communication Skills	Job Offer
≥8	Yes	Good	Good	Yes
≥8	Yes	Average	Good	Yes

Working of Naïve Bayes

Step 1

Collect training dataset.

Example:

Student data with CGPA, communication skills, etc.

Step 2

Calculate prior probabilities.

Example:

Probability of Yes

$$P(Yes) = \frac{7}{10} = 0.7$$

Probability of No

$$P(No) = \frac{3}{10} = 0.3$$

Step 3

Calculate likelihood probabilities for each attribute.

Calculate Likelihoods for YES

We calculate probabilities using only rows where Job Offer = Yes.

Total Yes rows = 7

1. CGPA ≥8 given Yes

All 7 Yes rows have CGPA ≥8.

$$P(CGPA \geq 8|Yes) = \frac{7}{7} = 1$$

2. Interactiveness = Yes given Yes

"Yes" appears 5 times.

$$P(Interactive = Yes|Yes) = \frac{5}{7}$$

3. Practical Knowledge = Good given Yes

"Good" appears 4 times.

$$P(Practical = Good|Yes) = \frac{4}{7}$$

4. Communication Skills = Good given Yes

"Good" appears 4 times.

$$P(Communication = Good|Yes) = \frac{4}{7}$$

Step 5: Calculate Posterior Probability for YES

Using Naïve Bayes:

$$P(Yes|X) = P(CGPA|Yes) \times P(Interactive|Yes) \times P(Practical|Yes) \times P(Communication|Yes) \times P(Yes)$$

Substitute values:

$$P(Yes|X) = 1 \times \frac{5}{7} \times \frac{4}{7} \times \frac{4}{7} \times 0.7$$

$$= 0.163$$

Step 6: Calculate Likelihoods for NO

Total No rows = 3

1. CGPA ≥8 given No

Appears 1 time.

$$P(CGPA \geq 8|No) = \frac{1}{3}$$

2. Interactiveness = Yes given No

Appears 1 time.

$$P(Interactive = Yes|No) = \frac{1}{3}$$

3. Practical Knowledge = Good given No

Appears 1 time.

$$P(Practical = Good|No) = \frac{1}{3}$$

4. Communication Skills = Good given No

Appears 1 time.

$$P(Communication = Good|No) = \frac{1}{3}$$

Step 7: Calculate Posterior Probability for NO

$$P(No|X) = P(CGPA|No) \times P(Interactive|No) \times P(Practical|No) \times P(Communication|No) \times P(No)$$

Substitute values:

$$P(No|X) = \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} \times 0.3$$

$$= 0.0037$$

Step 8: Compare Probabilities

Class	Probability
Yes	0.163
No	0.0037

Since:

0.163>0.0037

Final Prediction

✅ **Job Offer = YES**

The student with following attributes is likely to get a job offer.

Attribute	Value
CGPA	≥8
Interactiveness	Yes
Practical Knowledge	Good
Communication Skills	Good

Clustering Algorithms

Definition

Clustering is the process of dividing a dataset into groups (clusters) such that:

- Objects within the same cluster are similar.
- Objects in different clusters are dissimilar.
- No class labels are provided beforehand.

Need for Clustering

Clustering helps in:

- Customer segmentation
- Image segmentation
- Medical diagnosis
- Gene analysis
- Pattern recognition
- Market analysis

Characteristics of Clustering

1. Unsupervised learning technique.
2. Uses unlabeled data.
3. Groups similar objects together.
4. Helps discover hidden patterns in data.
5. Reduces complexity of large datasets.

Clustering Process

- Collect unlabeled data.
- Measure similarity between data objects.
- Apply clustering algorithm.
- Form clusters.
- Analyze and interpret clusters.

Types of Clustering Algorithms

1. K-Means Clustering

K-Means is the most popular partitioning clustering algorithm.

Steps

1. Select K cluster centers (centroids).
2. Assign each data point to the nearest centroid.
3. Recalculate centroids.
4. Repeat until centroids stop changing.

Advantages

- Simple and fast.
- Efficient for large datasets.
- Easy to implement.

Disadvantages

- Need to specify K beforehand.
- Sensitive to outliers.
- May converge to local optimum.

Example

Suppose student marks:

10, 12, 15, 80, 85, 90

For K=2:

- Cluster 1 → {10, 12, 15}
- Cluster 2 → {80, 85, 90}

2. Hierarchical Clustering

Creates a hierarchy of clusters represented as a tree structure called a **Dendrogram**.

Types

Agglomerative (Bottom-Up)

- Each object starts as its own cluster.
- Closest clusters are merged repeatedly.
- Continues until one cluster remains.

Divisive (Top-Down)

- Starts with one large cluster.
- Splits clusters recursively.

Advantages

- No need to specify number of clusters initially.
- Produces dendrogram for visualization.

Disadvantages

- Computationally expensive.
- Difficult for very large datasets.

Difference Between K-Means and Hierarchical Clustering

Feature	K-Means	Hierarchical
Type	Partitioning	Tree-based
Number of clusters	Required beforehand	Not required initially
Speed	Faster	Slower
Large datasets	Suitable	Less suitable
Output	Clusters	Dendrogram

Applications of Clustering

- Customer grouping in marketing
- Social network analysis
- Image processing
- Medical diagnosis
- Document classification
- Pattern recognition
- Recommendation systems

Advantages of Clustering

- Discovers hidden patterns.
- Works with unlabeled data.
- Useful for exploratory data analysis.
- Reduces data complexity.

Limitations of Clustering

- Results depend on similarity measures.
- Difficult to determine optimal number of clusters.
- Sensitive to noise and outliers.
- Different algorithms may produce different clusters.

Clustering Analysis is an unsupervised learning technique that groups similar objects into clusters. Objects within a cluster are highly similar and objects in different clusters are dissimilar. Common clustering algorithms are **K-Means** and **Hierarchical Clustering**. Clustering is widely used in market segmentation, image processing, pattern recognition, medical diagnosis, and recommendation systems. It helps discover hidden patterns in unlabeled data.

Clustering Analysis is a type of unsupervised learning used to divide data into groups called clusters. It aims to place similar objects in the same cluster and dissimilar objects in different clusters.

Types of Clustering Algorithms

1. K-Means Clustering
2. Hierarchical Clustering

K-Means Algorithm

- Choose K centroids.
- Assign points to nearest centroid.
- Recalculate centroids.
- Repeat until convergence.

Hierarchical Clustering

- Agglomerative (Bottom-Up)
- Divisive (Top-Down)
- Represented using a dendrogram.

Applications

- Customer segmentation
- Image segmentation
- Pattern recognition
- Medical diagnosis
- Recommendation systems

Advantages

- Finds hidden patterns.
- No labeled data required.
- Useful for exploratory analysis.

Limitations

- Sensitive to outliers.
- Choosing optimal clusters is difficult.
- Computationally expensive for large datasets.

Thus, clustering is an important unsupervised learning technique for discovering meaningful groups in data.

Differences Between Clustering and Classification

Clustering	**Classification**
Unsupervised learning	Supervised learning
No supervisor available	Supervisor available
Uses unlabelled data	Uses labelled data
No prior knowledge required	Domain knowledge required
Cluster results are dynamic	Labels usually remain fixed

Applications of Clustering

1. Grouping customers based on buying patterns.
2. Customer profiling based on lifestyle.
3. Information retrieval (finding documents from document collections).
4. Identifying groups of genes related to diseases.
5. Identifying organs with similar physiological functions.
6. Taxonomy of plants and animals in Biology.
7. Clustering based on purchasing behaviour and demographics.
8. Document indexing.
9. Data compression by grouping similar objects and identifying duplicates.

Challenges of Clustering Algorithms

1. High-Dimensional Data

- Modern datasets contain many features.
- Clustering becomes difficult when dimensions increase.

2. Large Volume of Data

- Internet generates billions of data records.
- Processing such huge datasets is challenging.

3. Scalability Problem

- Some algorithms work well on small datasets.
- The same algorithms may perform poorly on large datasets.

4. Different Units of Measurement

Example:

- Weight in kilograms (kg)
- Weight in pounds (lb)

Different units can affect clustering results.

5. Proximity Measure Design

- Choosing the correct similarity or distance measure is difficult.
- A poor distance measure leads to poor clusters.

Proximity Measures

What are Proximity Measures?

Clustering algorithms need a way to find how similar or different objects are so that they can be grouped into clusters.

- **Similarity** = How much two objects are alike.
- **Dissimilarity** = How much two objects are different.

These together are called **Proximity Measures**.

Usually, **distance measures** are used to find similarity between two objects i and j.

Distance Measures

Distance measures are also called **dissimilarity measures** because they indicate how different one object is from another.

Example:

- Smaller distance → More similar
- Larger distance → More different

Some measures such as **Cosine Similarity** directly indicate similarity between objects.

Distance between two objects i and j is denoted by:

$$D_{ij}$$

Properties of Distance Measures

1. Non-Negativity

$$D_{ij} \geq 0$$

- Distance is always positive or zero.
- Distance can never be negative.

2. Identity Property

$$D_{ii} = 0$$

3. Symmetry Property

$$D_{ij} = D_{ji}$$

Distance from object i to j is the same as distance from j to i

Example:

Distance(A,B) = Distance(B,A)

4. Triangle Inequality

$$D_{ij} \leq D_{ik} + D_{kj}$$

- Direct distance between two objects should be less than or equal to the distance through another object.

Example:

$A \rightarrow B \leq A \rightarrow C + C \rightarrow B$

Metric Distance

If all four properties are satisfied, then the distance measure is called a **Metric**.

Distance Measures Based on Data Types

Distance measures depend on the type of data used.

Data types are mainly divided into:

1. Categorical Variables

Values represent categories.

Example:

Gender = {Male, Female}

These are called **Nominal Variables**.

---

2. Ordinal Variables

These are similar to nominal variables but have an inherent order.

Example:

Temperature = {Low, Medium, High}

Order exists:

$$Low < Medium < High$$

3. Quantitative Variables

These contain numerical values.

Examples:

- Real numbers
- Integer numbers

Example:

- Age = 20
- Salary = 30000

4. Binary Data

Attributes can take only two values.

Example:

- Yes / No
- True / False
- 1 / 0

Objects having binary attributes are called Binary Objects.

Quantitative Variables

Quantitative variables are numerical variables. Distance measures are commonly used for these variables.

1. Euclidean Distance

- Euclidean distance is the most common distance measure.
- It is also called **L_2 norm**.
- It is the square root of the sum of squared differences between coordinates of two objects.

Formula

$$D(x_i, x_j) = \sqrt{\sum_{k=1}^{n}(x_{ik} - x_{jk})^2}$$

Advantages

- Distance does not change when new objects are added.

Disadvantages

- If units change, the distance changes significantly.
- Computational cost is high because of square and square root operations.

2. City Block Distance (Manhattan Distance)

Also called:

- Manhattan Distance
- Taxicab Distance

- Boxcar Distance
- Absolute Value Distance
- L_1 norm

Formula

$$D(x_i, x_j) = \sum_{k=1}^{n} |x_{ik} - x_{jk}|$$

- Distance is calculated by adding the absolute differences of coordinates.

3. Chebyshev Distance

Also called:

- Maximum Value Distance
- Supremum Distance
- $L\infty$ norm

Formula

$$D(x_i, x_j) = \max_k |x_{ik} - x_{jk}|$$

Take the maximum difference among all coordinates.

Partitional Clustering Algorithm (K-Means)

Introduction

- **K-means** is a simple and iterative partitional clustering algorithm.
- Here, **k** represents the number of clusters specified by the user.
- The algorithm assumes that clusters do **not overlap**.
- A data point can belong to **only one cluster**.
- It can detect clusters having **circular or spherical shapes**.

Initialization

- The algorithm starts by selecting **k data points** randomly as initial cluster centers (centroids).
- Prior knowledge is usually absent.
- Therefore, choosing good initial centroids is important.
- Data points should be **normalized** for better performance.

Working of K-Means

The main objective is to assign each data point to the nearest cluster centroid.

The process continues until no further changes occur in cluster assignments.

Steps of K-Means Algorithm

Step 1

Determine the number of clusters (**k**).

Step 2

Choose **k instances randomly** as initial cluster centers.

Step 3

Compute the centroid and assign each sample to the nearest cluster using Euclidean distance.

Step 4

Re-compute the centroid using newly assigned samples.

Step 5

Repeat Steps 3 and 4 until no changes occur in cluster assignments.

Sum of Squared Error (SSE)

K-means attempts to minimize the **Sum of Squared Error (SSE)**.

$$SSE = \sum_{i=1}^{k} dist(c_i, x)^2$$

$SSE = \sum_{i=1}^{k} dist(c_i, x)^2$

Where:

- c_i = centroid of cluster i
- x = data point
- $dist$ = Euclidean distance

Aim

Minimize SSE.

Advantages

1. Simple

2. Easy to implement

Disadvantages

1. Sensitive to initialization; different starting points may produce different clusters.
2. For large datasets, execution time increases.

How to Choose the Value of k?

- There is no fixed rule for selecting **k**.
- Run K-means for different values of **k**.
- Compute within-group variance (sum of squares).
- Plot the results as a graph called the **Elbow Curve**.
- The point where the graph becomes almost flat indicates the optimal value of **k**.

Complexity

The complexity depends on:

- n = number of samples
- k = number of clusters
- I = number of iterations
- d = number of attributes

$$\Theta(nkId)$$

The complexity of K-means algorithm is:

$$O(n^2)$$

Example 13.5: Consider the following set of data given in Table 13.9. Cluster it using *k*-means algorithm with the initial value of objects 2 and 5 with the coordinate values (4, 6) and (12, 4) as initial seeds.

Table 13.9: Sample Data

Objects	X-coordinate	Y-coordinate
1	2	4
2	4	6
3	6	8
4	10	4
5	12	4

Initial Seeds

Object 2 = (4,6)

Object 5 = (12,4)

Thus:

Cluster 1 Centroid (4,6)

Cluster 2 Centroid (12,4)

Iteration 1

Object 1 = (2,4)

Distance to Centroid 1:

$$\sqrt{(2-4)^2+(4-6)^2}$$
$$=\sqrt{8}$$

Distance to Centroid 2:

$$\sqrt{(2-12)^2+(4-4)^2}$$
$$=\sqrt{100}$$
$$=10$$

Therefore, Object 1 belongs to Cluster 1.

Object 3 = (6,8)

Distance to Centroid 1:

$$\sqrt{(6-4)^2+(8-6)^2}$$
$$=\sqrt{8}$$

Distance to Centroid 2:

$$\sqrt{(6-12)^2+(8-4)^2}$$
$$=\sqrt{52}$$

Object 3 belongs to Cluster 1.

Object 4 = (10,4)

Distance to Centroid 1:

$$\sqrt{(10-4)^2+(4-6)^2}$$
$$=\sqrt{40}$$

Distance to Centroid 2:

$$\sqrt{(10-12)^2+(4-4)^2}$$
$$=\sqrt{4}$$
$$=2$$

Object 4 belongs to Cluster 2.

Object 5 = (12,4)

Already belongs to Cluster 2.

Cluster Table After Iteration 1

Cluster 1

- (4,6)
- (2,4)
- (6,8)

Cluster 2

- (10,4)
- (12,4)

New Centroids

Cluster 1

$$\left(\frac{4+2+6}{3}, \frac{6+4+8}{3}\right)$$

$$= (4, 6)$$

Cluster 2

$$\left(\frac{10+12}{2}, \frac{4+4}{2}\right)$$

$$= (11, 4)$$

Iteration 2

Object 1 (2,4)

Distance to Cluster 1:

$$\sqrt{8}$$

Distance to Cluster 2:

$$\sqrt{81}$$
$$= 9$$

Remains in Cluster 1.

Object 3 (6,8)

Distance to Cluster 1:

$$\sqrt{8}$$

Distance to Cluster 2:

$$\sqrt{41}$$

Remains in Cluster 1.

Object 4 (10,4)

Distance to Cluster 1:

$$\sqrt{40}$$

Distance to Cluster 2:

$$\sqrt{1}$$

$$= 1$$

Remains in Cluster 2.

Object 5 (12,4)

Distance to Cluster 1:

$$\sqrt{68}$$

Distance to Cluster 2:

$$\sqrt{1}$$

$$= 1$$

Remains in Cluster 2.

Final Cluster Table

Cluster 1

- (4,6)
- (2,4)
- (6,8)

Centroid = (4,6)

Cluster 2

- (10,4)
- (12,4)

Centroid = (11,4)

Result

No change occurs in the second iteration.

Therefore, the **K-means algorithm terminates**.

Final Clusters

Cluster 1: {(2,4), (4,6), (6,8)}

Cluster 2: {(10,4), (12,4)}

Example 8.4

Assess a Student's performance. Using the Naive Bayes algorithm for continuous attributes.

Predict whether a student get a job offer or not in the final year using the training dataset T consisting of 10 data instances with attributes such as "CGPA" and "Interactiveness" shown in Table 8.13. The target variable is job offer, which is classified as yes or No. for a Student.

Training Dataset

Sl No	CGPA	Interactiveness	Job offer
1	9.5	yes	yes
2	8.2	No	yes
3	9.3	No	NO
4	7.6	No	No
5	8.4	yes	yes
6	9.1	yes	yes
7	7.5	yes	NO
8	9.6	No	yes
9	8.6	yes	yes
10	8.3	yes	yes

Test Data

Predict the job offer for a student having: CGPA = 8.5 and Interactiveness = yes using Gaussian Naive Bayes.

Step 1: Prior Probability

From the table:

Job offer	count
Yes	7
NO	3

Therefore:

$$P(\text{JobOffer} = \text{Yes}) = \frac{7}{10}$$

$$P(\text{Job offer} = \text{No}) = \frac{3}{10}$$

Step 2: Interactiveness probabilities.

For Job offer = yes

Out of 7 yes records:

* Interactive = Yes → 5

Therefore:

$$P(\text{Interactive} = \text{yes} \mid \text{Job Offer} = \text{yes}) = \frac{5}{7}$$

for Job offer = No

Out of 3 No records

* Interactive = yes → 1

Therefore:

$$P(\text{Interactive} = \text{yes} \mid \text{Job offer} = \text{No}) = \frac{1}{3}$$

Step 3: Continuous Attributes (CGPA)

For yes class,

1. Find μ_{yes} (Mean for Job offer = yes)

Take only the CGPA values where Job offer = yes:

CGPA

9.5

8.2

8.4

9.1

9.6

8.6

8.3

formula

$$\mu = \frac{\Sigma x}{N}$$

Adding all values: $9.5 + 8.2 + 8.4 + 9.1 + 9.6 + 8.6 + 8.3$

$= 61.7$

No of values $= 7$

$$\mu_{yes} = \frac{61.7}{7} = 8.814286$$

Q Find σ_{yes} (Standard Deviation)

Formula

$$\sigma = \sqrt{\frac{\Sigma (x-\mu)^2}{N}}$$

Using $\mu = 8.8143$

x	$x-\mu$	$(x-\mu)^2$
9.5	0.6857	0.4702
8.2	−0.6143	0.3774
8.4	−0.4143	0.1716
9.1	0.2857	0.0816
9.6	0.7857	0.6173
8.6	−0.2143	0.0459
8.3	−0.5143	0.2645

Add the last column: 2.0285

Variance: $\frac{2.0285}{7} = 0.2898$

Standard deviation:

$$\sigma_{yes} = \sqrt{0.2898}$$

$$= 0.5383$$

3. Find μ_{NO}.

Take only Job Offer = No:

CGPA

9.3

7.6

7.5

Add: $9.3 + 7.6 + 7.5 = 24.4$

Divide by 3:

$$\mu_{NO} = \frac{24.4}{3} = 8.1333$$

4. Find σ_{NO}.

Using $\mu = 8.1333$

x	$x - \mu$	$(x-\mu)^2$
9.3	1.1667	1.3611
7.6	-0.5333	0.2844
7.5	-0.6333	0.4011

Add Squares $1.3611 + 0.2844 + 0.4011 = 2.0466$

$$\text{variance} = \frac{2.0466}{3} = 0.6822$$

Standard deviation:

$$\sigma_{NO} = \sqrt{0.6822}$$

$$= 0.826$$

Step 4: Calculate Gaussian probability

Test CGPA:

$x = 8.5$

Use Gaussian formula:

$$P(x = x/C) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

For yes

Substitute $x = 8.5$

$\mu = 8.8714$

$\sigma = 0.38146$

$$P(CGPA = 8.5 \mid yes) = 0.597$$

For No.

Substitute $x = 8.5$

$\mu = 8.1333$

$\sigma = 1.0116$

$$P(CGPA = 8.5 \mid No) = 0.369$$

Step 5: Apply Naive Bayes

For YES

$$P(yes) \times P(CGPA/yes) \times P(Interactive = yes)$$

$$= \frac{7}{10} \times 0.594 \times \frac{5}{7}$$

$$= 0.297$$

$$=$$

Step 6: for NO

$$P(No) \times P(CGPA|No) \times P(Interactive=Yes|No)$$

$$= \frac{3}{10} \times 0.369 \times \frac{1}{3}$$

$$= 0.0369$$

Step 7: Compare

class	probability
Yes	0.297
No	0.0369

Since $0.297 > 0.0369$

Final Answer

Job Offer = Yes

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Module 5
Chapter 3: Artificial Neural Networks (ANN)

Introduction

Artificial Neural Networks (ANNs) are computational models inspired by the structure and functioning of the human brain.They consist of interconnected processing units called neurons that learn patterns from data.ANNs are capable of classification, prediction, pattern recognition, speech processing, image analysis, medical diagnosis, recommendation systems, and forecasting.

Characteristics:
• Learning from examples
• Adaptive behavior
• Parallel processing
• Fault tolerance
• Generalization capability

Biological Neurons
A biological neuron is the fundamental processing unit of the nervous system.

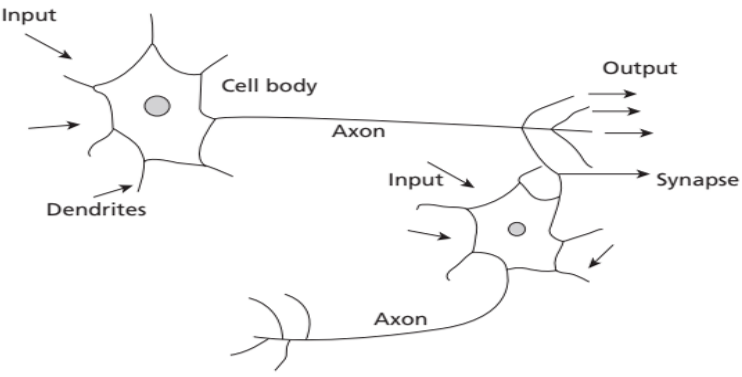


Figure 10.1: A Biological Neuron

Functions:

• Dendrites receive signals.

• Soma integrates information.

• Axon transmits impulses.

• Synapses connect neurons.

The brain contains billions of interconnected neurons forming a highly complex processing network.

Artificial Neuron Model

Artificial neurons mimic the behavior of biological neurons.

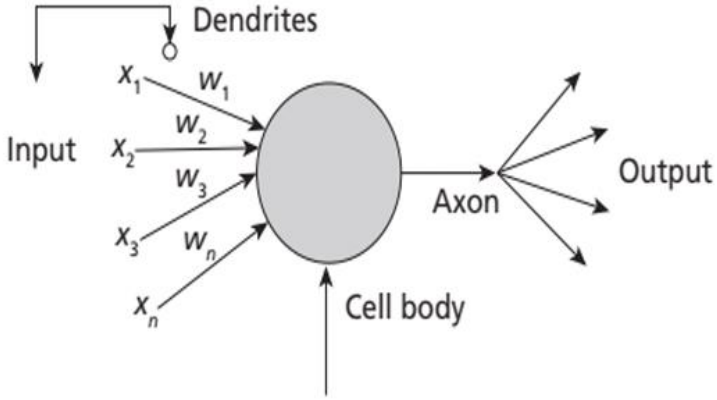


Figure 10.2: An Artificial Neuron

Operation:

Net Input = $\Sigma(w_i x_i) + b$

where:

xi = inputs
wi = weights
b = bias

The activation function transforms the weighted sum into an output signal.

ANN Structure
An ANN is organized into layers.

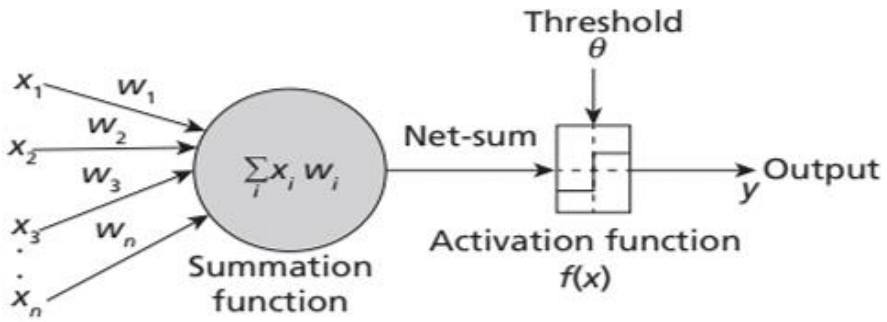


Figure 10.3: McCulloch & Pitts Neuron Mathematical Model

Components of a Perceptron

The Perceptron is the simplest form of an Artificial Neural Network (ANN). It consists of four major components:

1. Inputs
2. Weights and Bias
3. Net Sum (Summation Function)
4. Activation Function

1. **Inputs**: are the signals or data values received by a neuron from the external environment or from other neurons. They represent the features or attributes of a problem.

   Explanation: Inputs are denoted by x_1, x_2, x_3, ..., x_n. Each input corresponds to a particular characteristic of the data being processed.

   The input layer of a neural network receives these values and forwards them to the neuron for further processing.

Importance

- Inputs provide information to the neuron.
- The quality of inputs directly affects the performance of the neural network.
- More relevant inputs generally lead to better learning and prediction.

2. Weights and Bias:

Weights are numerical values assigned to inputs that indicate their importance in the decision-making process. Bias is an additional parameter used to shift the activation function and improve learning.

Each input is associated with a weight.

$w_1, w_2, w_3, \ldots, w_n$

A weight determines how strongly an input influences the output.

Bias

Bias is an additional value added to the weighted sum before applying the activation function.

Purpose of Bias:

- Helps shift the decision boundary.
- Allows the neuron to activate even when all inputs are zero.
- Improves flexibility and learning capability.

Importance

- Weights determine the importance of each feature.
- Bias improves the accuracy of classification.
- Learning in neural networks mainly involves adjusting weights and bias values.

3. Net Sum (Summation Function)

Definition

The Net Sum is the weighted combination of all inputs received by a neuron. It is calculated by multiplying each input by its corresponding weight and then adding the results together along with the bias.

Mathematical Expression

Net=∑i=1nwixi+bNet=\sum_{i=1}^{n} w_i x_i+bNet=∑i=1nwixi+b

Explanation

The summation unit performs the following operations:

1. Multiply each input by its corresponding weight.
2. Add all weighted inputs.
3. Add the bias value.

The resulting value is called the Net Input or Net Sum.

4. Activation Function

Definition

An Activation Function is a mathematical function that determines whether a neuron should produce an output and what that output should be.

Explanation

After computing the Net Sum, the neuron must decide whether it should be activated (fire) or not. This decision is made by the activation function.

The activation function transforms the net input into an output value.

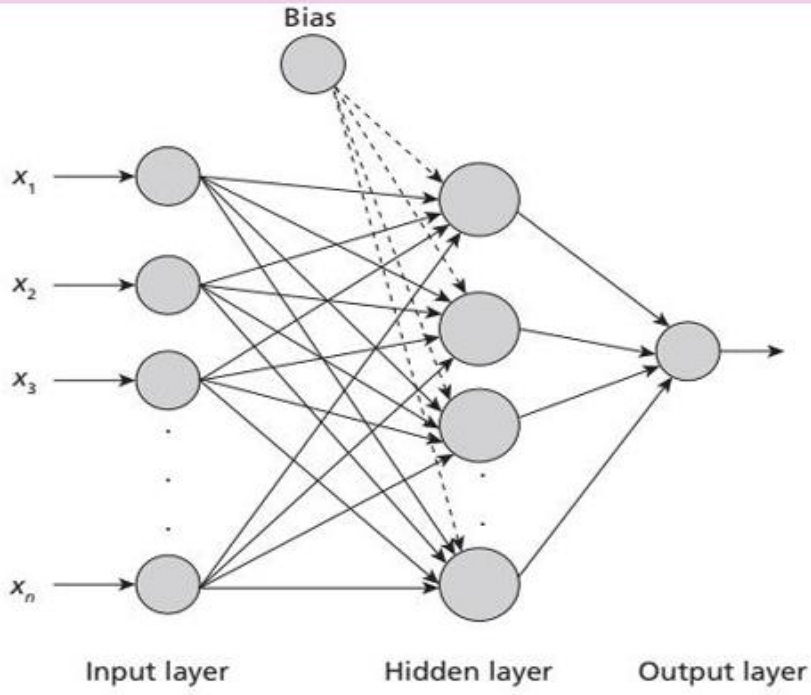


Figure 10.4: Artificial Neural Network Structure

Activation Functions

Activation functions determine the output of a neuron and introduce non-linearity into the network.

Why Activation Functions are Needed?

Without activation functions, a neural network behaves like a simple linear model regardless of the number of layers.

Activation functions allow neural networks to learn complex and non-linear relationships.

1. Identity Function or Linear Function

$$f(x) = x \ \forall x \tag{10.4}$$

The value of $f(x)$ increases linearly or proportionally with the value of x. This function is useful when we do not want to apply any threshold. The output would be just the weighted sum of input values. The output value ranges between $-\infty$ and $+\infty$.

2. Binary Step Function

$$f(x) = \begin{cases} 1 & if\ f(x) \geq \theta \\ 0 & if\ f(x) < \theta \end{cases} \tag{10.5}$$

The output value is binary, i.e., 0 or 1 based on the threshold value θ. If value of $f(x)$ is greater than or equal to θ, it outputs 1 or else it outputs 0.

3. Bipolar Step Function

$$f(x) = \begin{cases} 1 & if\ f(x) \geq \theta \\ -1 & if\ f(x) < \theta \end{cases} \tag{10.6}$$

The output value is bipolar, i.e., +1 or −1 based on the threshold value θ. If value of $f(x)$ is greater than or equal to θ, it outputs +1 or else it outputs −1.

4. Sigmoidal Function or Logistic Function

$$\sigma(x) = \frac{1}{1 + e^{-x}} \tag{10.7}$$

It is a widely used non-linear activation function which produces an S-shaped curve and the output values are in the range of 0 and 1. It has a vanishing gradient problem, i.e., no change in the prediction for very low input values and very high input values.

5. Bipolar Sigmoid Function

$$\sigma(x) = \frac{1 - e^{-x}}{1 + e^{-x}} \tag{10.8}$$

It outputs values between −1 and +1.

6. Ramp Functions

$$f(x) = \begin{cases} 1 & if\ x > 1 \\ x & if\ 0 \leq x \leq 1 \\ 0 & if\ x < 0 \end{cases} \tag{10.9}$$

It is a linear function whose upper and lower limits are fixed.

Perceptron and Learning Theory

The perceptron is the earliest neural network model proposed by Frank Rosenblatt.

Architecture:

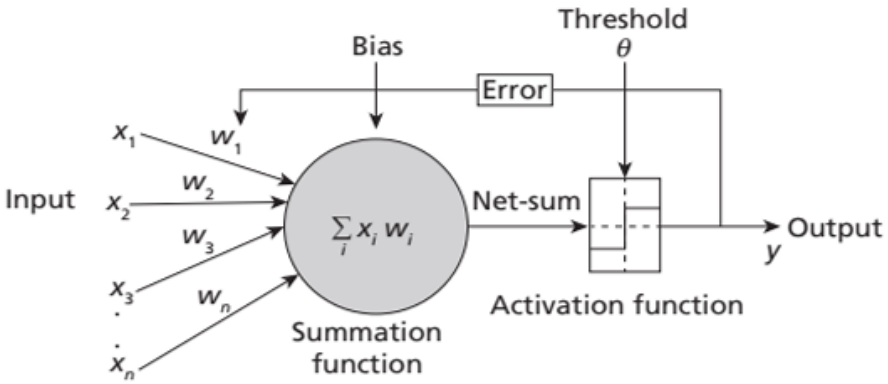


Figure 10.5: Perceptron Model

Characteristics:
• Single-layer network
• Supervised learning
• Binary classification

Learning Theory:
The perceptron learns by adjusting weights to reduce prediction errors.
The objective is to correctly classify training examples.

Limitations:
• Can only solve linearly separable problems.
• Cannot solve XOR problems.

Perceptron Algorithm

Algorithm 10.1: Perceptron Algorithm

Set initial weights $w_1, w_2, \ldots, w_n$ and bias θ to a random value in the range [–0.5, 0.5].

For each Epoch,

1. Compute the weighted sum by multiplying the inputs with the weights and add the products.
2. Apply the activation function on the weighted sum:
$$Y = \text{Step}((x_1 w_1 + x_2 w_2) - \theta)$$
3. If the sum is above the threshold value, output the value as positive else output the value as negative.
4. Calculate the error by subtracting the estimated output $Y_{estimated}$ from the desired output $Y_{desired}$:
$$\text{error } e(t) = Y_{desired} - Y_{estimated}$$
[If error $e(t)$ is positive, increase the perceptron output Y and if it is negative, decrease the perceptron output Y.]
5. Update the weights if there is an error:
$$\Delta w_i = \propto \times e(t) \times x_i$$
$$w_i = w_i + \Delta w_i$$
where, x_i is the input value, $e(t)$ is the error at step t, $\propto$ is the learning rate and Δw_i is the difference in weight that has to be added to w_i.

Advantages:

• Simple implementation.

• Fast convergence for linearly separable data.

Disadvantages:

• Limited learning capability.

Types of Artificial Neural Networks

Feed Forward Neural Network (FFNN) : A Feed Forward Neural Network is the simplest type of neural network in which information flows only in one direction, from the input layer to the output layer through one or more hidden layers. There are no feedback connections or loops.

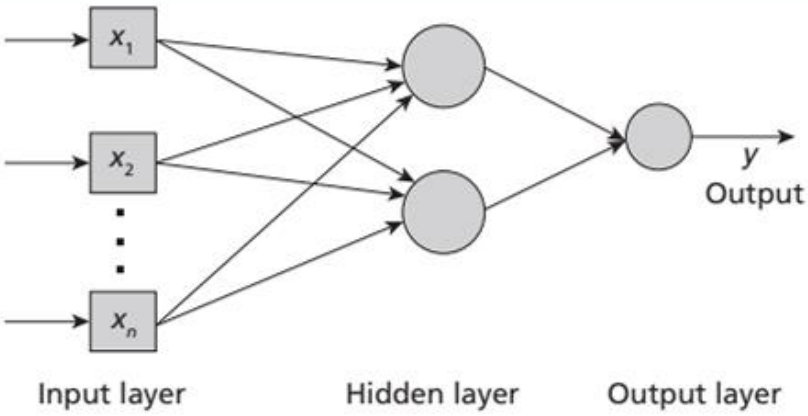


Figure 10.7: Model of a Feed Forward Neural Network

Characteristics

- **One-way flow of information.**
- **No memory of previous inputs.**
- **Simple architecture.**
- **Easy to train and implement.**

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2. Fully Connected Neural Network (FCNN)

In a Fully Connected Neural Network, every neuron in one layer is connected to every neuron in the next layer. This allows maximum information transfer between layers.

Characteristics

- Dense connections between layers.
- Every neuron participates in learning.
- Large number of trainable parameters.

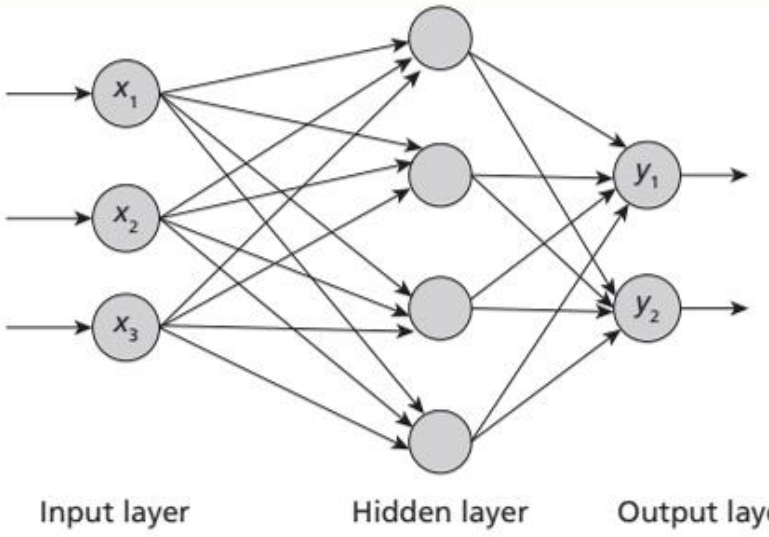


Figure 10.8: Model of a Fully Connected Neural Network

3. Multi-Layer Perceptron (MLP)

A Multi-Layer Perceptron is an extension of the perceptron that contains one or more hidden layers between the input and output layers. It uses non-linear activation functions and is trained using the Backpropagation algorithm.

Characteristics

- Contains multiple hidden layers.
- Supports supervised learning.
- Can solve non-linearly separable problems.
- Uses activation functions such as Sigmoid, Tanh, or ReLU.

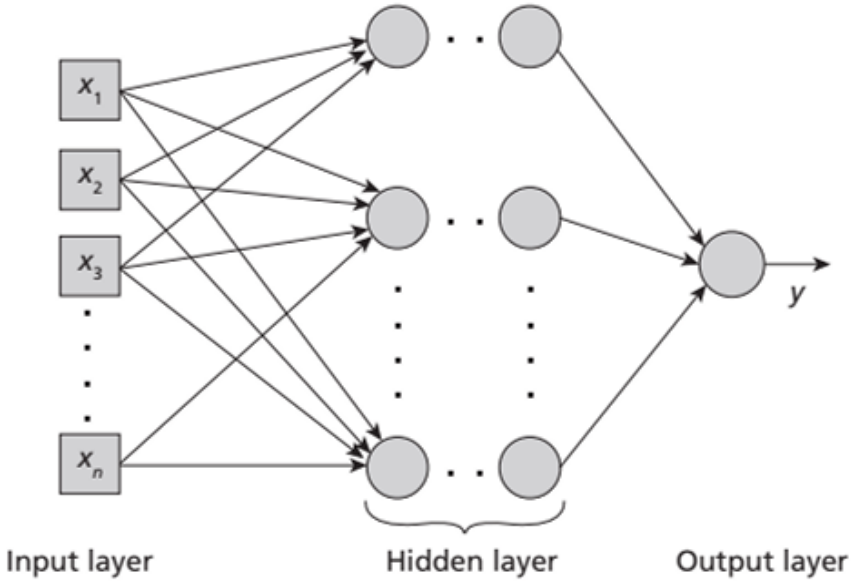


Figure 10.9: Model of a Multi-Layer Perceptron

4. Feedback Neural Network (Recurrent Neural Network)

A Feedback Neural Network contains feedback connections where the output of a neuron can be fed back into the network as input. This enables the network to remember previous information and process sequential data.

Characteristics

- Contains loops or feedback paths.
- Has memory of previous inputs.
- Information flows in both forward and backward directions.
- Suitable for temporal and sequential data.

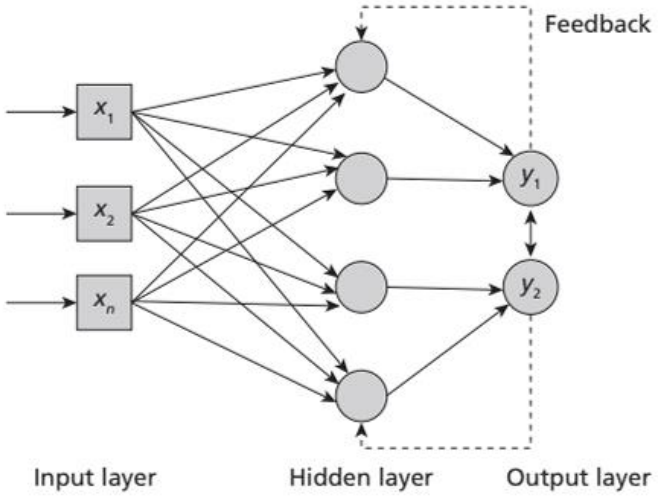


Figure 10.10: Model of a Feedback Neural Network